Marvell Technology (MRVL)

Full investment research | Model assumptions and source evidence

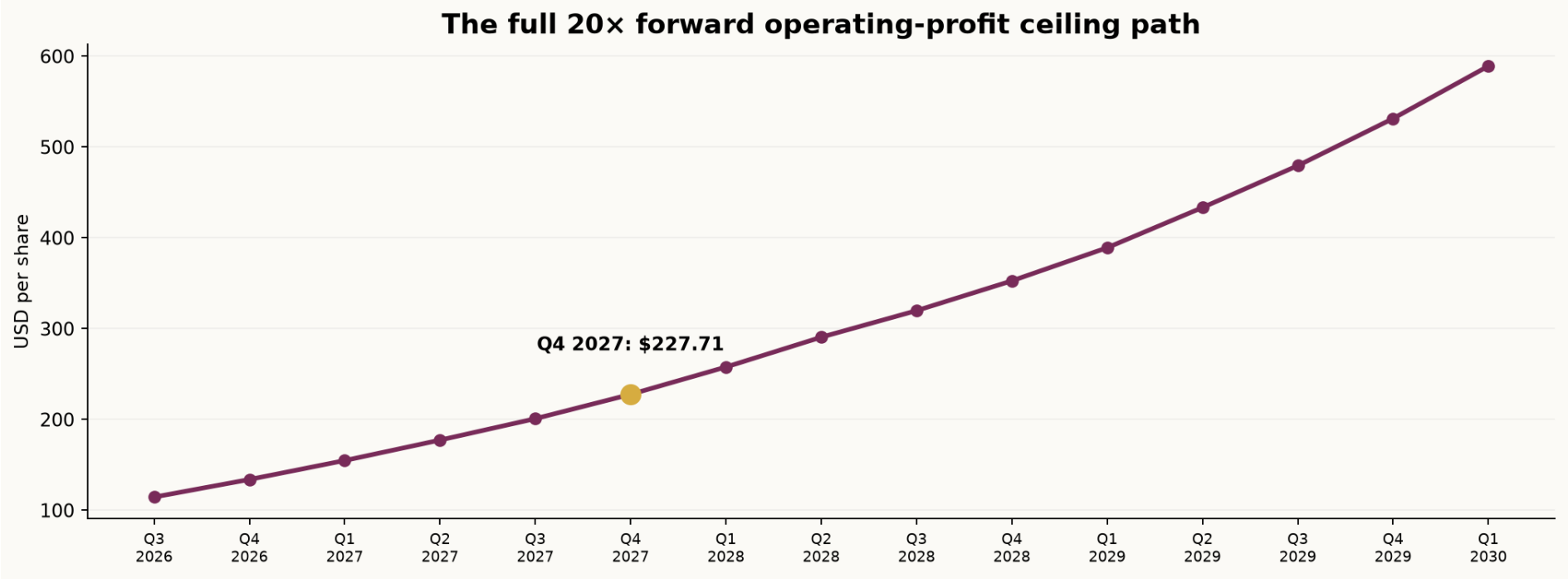
Current full research artifact | Evidence update 12 October 2026 | Framework 8.0
Investor Day transcript, Q&A and all 161 supplied deck pages reviewed. Original slides are embedded beside the relevant analysis. USD unless stated; deck reconciliation completed October 6, 2026. The October 12 CXL product evidence is added below with no change to operating forecasts, targets, valuation method or scores.

Q4 2027 model ceiling: $227.71 per share

Marvell's growth case is increasingly about the value of moving data. Management's FY2028 revenue outlook is approximately $20B, supported by stronger optics and switching demand. The active model includes that near-term guidance and a 70% weight on the incremental longer-term opportunity, with packaging costs and growth spending carried into profit. This is a model ceiling, not a forecast of the market's trading price.

Current model item	Value
Active horizon	Q4 2027
Price ceiling, USD/share	227.7064
Valuation basis	20× forward-four-quarter non-GAAP operating profit
Exact operating window	Q4 2027 through Q3 2028 (FQ3 FY2028 through FQ2 FY2029)
Forward operating profit, USD M	10,594.047
Target-quarter share proxy, M	930.501
Artifact analytical Base	85/100 (51/60 points)
Protected portfolio Excel Base	85/100

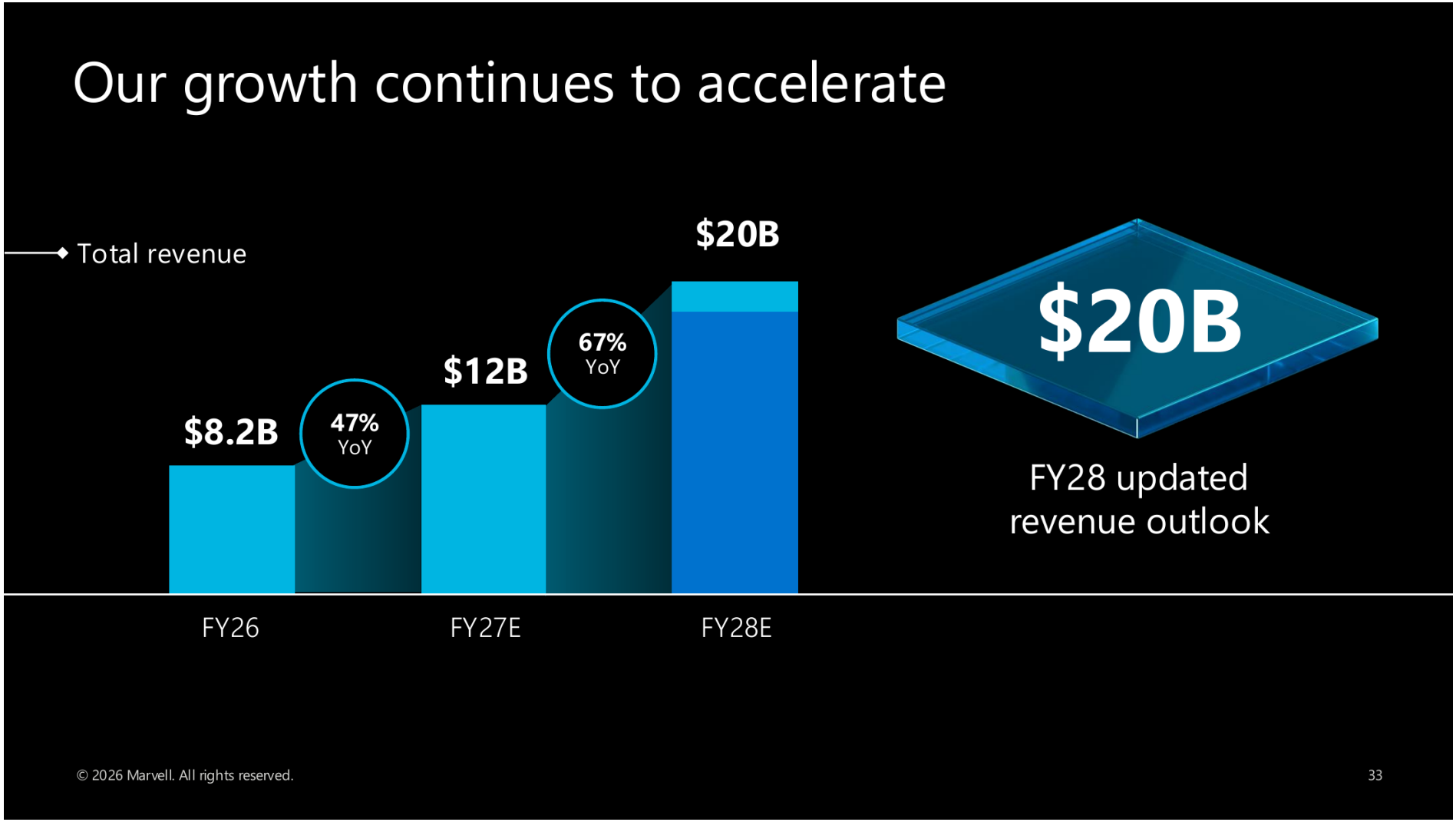
The denominator is the displayed ceiling quarter’s share proxy. The four-quarter window includes that quarter. No multiple expansion, after-tax EPS substitution, cash/debt adjustment or new share-buyback assumption is included. The current price input in Excel is retained. The October 7 custom-growth review changes the interpretation and watch conditions; it adds no second financial increment to the already adopted Investor Day forecast.



03 ceiling path

Ceiling quarter	Forward 4Q OP, USD M	Shares, M	20× ceiling, USD/share
Q3 2026	5,278.114	921.200	114.5921
Q4 2026	6,169.151	921.000	133.9664
Q1 2027	7,148.889	923.340	154.8484
Q2 2027	8,208.635	925.680	177.3535
Q3 2027	9,325.738	928.261	200.9293
Q4 2027	10,594.047	930.501	227.7064
Q1 2028	12,023.235	932.741	257.8044
Q2 2028	13,672.088	940.821	290.6417
Q3 2028	15,178.076	948.901	319.9086
Q4 2028	16,884.729	957.461	352.6980

Ceiling quarter	Forward 4Q OP, USD M	Shares, M	20× ceiling, USD/share
Q1 2029	18,797.648	966.021	389.1767
Q2 2029	20,998.671	968.981	433.4174
Q3 2029	23,309.409	972.182	479.5278
Q4 2029	25,918.011	975.382	531.4434
Q1 2030	28,835.285	978.822	589.1834



Marvell Investor Day, PDF page 33. Updated FY2028 revenue outlook of $20B versus approximately $12B FY2027: 67% rounded growth. These are fiscal years and company outlooks, not calendar-year results. The final slide corrects the stale 50% label on page 32.

October 12 CXL evidence: more memory behind the existing controller pool

The current Excel Base is **85/100** and the artifact analytical Base is **85/100**. Earlier artifact wording that called the portfolio Base 99 was stale after the owner's October 7 removal of overrides. This edition aligns the labels to the current master; it does not change either score. The active Q4 2027 ceiling remains **USD 227.7063746694**, at the preserved 20x forward-four-quarter non-GAAP operating-profit method and existing share schedule.

Marvell's October 12 issuer release introduces second-generation Structera X 3406 and 3506. The new generation provides six DDR channels, compared with four previously. It can expand memory behind compatible hosts and reuse DDR4 modules, improving useful memory capacity without implying new HBM supply. CXL protocol, host compatibility, software, latency, reliability and qualification determine which deployments create a paid Marvell controller invoice.

Disclosed capability	Exact scope and status
Six memory channels	50% more channels than the previous four-channel generation; not 50% more Marvell revenue.
Structera X 3406	Up to 18 DDR4 RDIMM slots and 9TB per controller.
Structera X 3506	Up to 12 DDR5 RDIMM slots and 6TB per controller.
Host connection	CXL 3.2 and PCIe 6.0 x16; up to 256GB/s host bandwidth, with memory sharing by up to four hosts.
Compression	Up to 250GB/s LZ4 compression/decompression; 2-3x effective memory capacity depends on the workload.
Vendor performance comparisons	Up to 2x DLRM queries per second, 40-50% faster time to first token at equivalent context length, or up to 2x supportable context at equivalent latency, versus the previous generation. These are issuer claims, not independently replicated system results.
Availability	Expected to sample in Q1 2027. First-generation 2404/2504 products are already shipping to hyperscalers. Sampling is not accepted production volume.

The issuer-distributed release was corrected at 13:25 EDT / 17:25 UTC to add a Meta ecosystem-support quote. That quote joins AMD, Arm, Alibaba, Samsung and SK hynix statements; it identifies collaboration and validation, without providing a specific controller volume, price, committed launch date or exclusive customer allocation. This correction is the same underlying product announcement, not an additional demand event.

Existing probability and overlap: the Investor Day opportunity retains its existing 70% weight on incremental FY2029-31 economics; FY2028 remains the full USD20B company-guidance anchor. Its USD70-90B FY2031 range and USD37.5B interconnect midpoint component were already applied on October 6 and reconciled on October 7. They are not counted again. The current merchant CXL/memory vector already includes controller adoption, DDR4 reuse and memory fabric: USD1.000B in FY2028, USD1.479571B in FY2029, USD2.174311B in FY2030 and USD3.067273B in FY2031, with a 63% product gross-margin allocation. Second-generation adoption can replace first-generation sales inside this pool. No fresh probability or additional controller-sales line is introduced by this evidence-only review.

Bounded company earnings bridge: USD100M of demonstrably incremental accepted controller invoices, after deducting replaced first-generation invoices and the existing merchant-memory trajectory, would carry USD37M of COGS and USD63M of GP at the retained product margin. Applying the existing FY2028 17% common expense ratio as a diagnostic would charge USD17M of operating costs, leaving USD46M of non-GAAP OP. This is a normalized cost transmission, not an adopted revenue forecast or a new controller ASP. Separate price, accepted unit count, net new versus replacement sales, launch engineering, support expense and qualification timing are not disclosed, so the source cannot populate a residual above the active pool. The zero added revenue, GP, opex and OP rests on this overlap and unquantified incremental invoice scope after the bounded attempt; it does not mean zero probability for product adoption or require official sales confirmation before credible forward evidence can matter.

What would change the operating model: a customer-specific accepted production ramp, net realized controller price, first-generation replacement rate, measured attach/utilization and incremental qualification/support cost. The same memory capacity cannot become both customer DRAM sales and Marvell controller sales. Compression can reduce new memory purchases for a fixed workload, while lower cost may support more workloads; these opposing mechanisms require separate supplier economics.

Sources and coverage: Marvell issuer release, October 12, 2026; direct retrieval returned HTTP403. The complete issuer-authored, corrected Business Wire release was recovered from its distributed reprint, cross-checked against the corrected second reprint. Business Wire source identity is preserved; direct binary retrieval disconnected. Grok OHSV R0 post 2109663976028352755 is a relay of this one event. Its mapped original image has not been delivered, so no image-only claim is inferred and it is not described as inspected. Existing original source images, the nine product vectors, all eighteen operating quarters and fifteen ceilings remain intact.

The investment thesis

Marvell sells the silicon and optical connections that keep expensive compute supplied with data. Its opportunity grows through more connectivity content per system, faster links and more deployed systems. Qualified designs, mixed-signal IP, software integration and the ability to supply both ends of a link create switching costs. The business has substantial competitors; it does not own the entire AI spending pool.

Scale-across connects data centers, scale-out connects systems, scale-up connects accelerators, and scale-in brings optics closer to chiplets and memory. CPO and NPO are integration choices, not additional TAMs. Qualified shipments drive the operating model; explicit probabilities admit credible evidence before all outcomes are known.

Business curve	Current stage	Economic test
Custom XPU and attach	Production with additional programs ramping	More sockets and net content per system must overcome wafer, packaging and customer-pricing costs.
Scale-out optics and switching	Established, accelerating production	1.6T mix and switch adoption lift sales; competition and more efficient network topologies limit automatic extrapolation.
Scale-up optics and switching	Qualification and early ramp	Customer acceptance, optical packaging yield and manufacturing capacity determine when high prospective content becomes revenue.
Scale-in optical I/O	Development and customer engagement	Useful demonstrations must become qualified, manufacturable systems; its eventual market is not near-term backlog.

Opportunity and the company capture boundary

Management's October 6 outlook is approximately USD12B FY2027 revenue and USD20B FY2028, including approximately USD18B of FY2028 data-center sales. The increase from the prior USD18B company outlook is led by scale-out optics, scale-up optics and switching. Management says demand and reserved supply support that plan. This is forward guidance, not an accepted-shipment total. [ID 51-57, 125]

For FY2031, management gives a USD70-90B company range, with a USD80B midpoint: USD37.5B interconnect, USD10B switching/storage, USD30B custom and USD2.5B other. The midpoint is a company forecast grounded in management's program assumptions, rather than a direct multiplication of capex. Customer forecasts,

architectural decisions and supply execution still govern realization. The probability ledger and operating model quantify the active outcome separately from these unweighted endpoints. [ID 59-73]

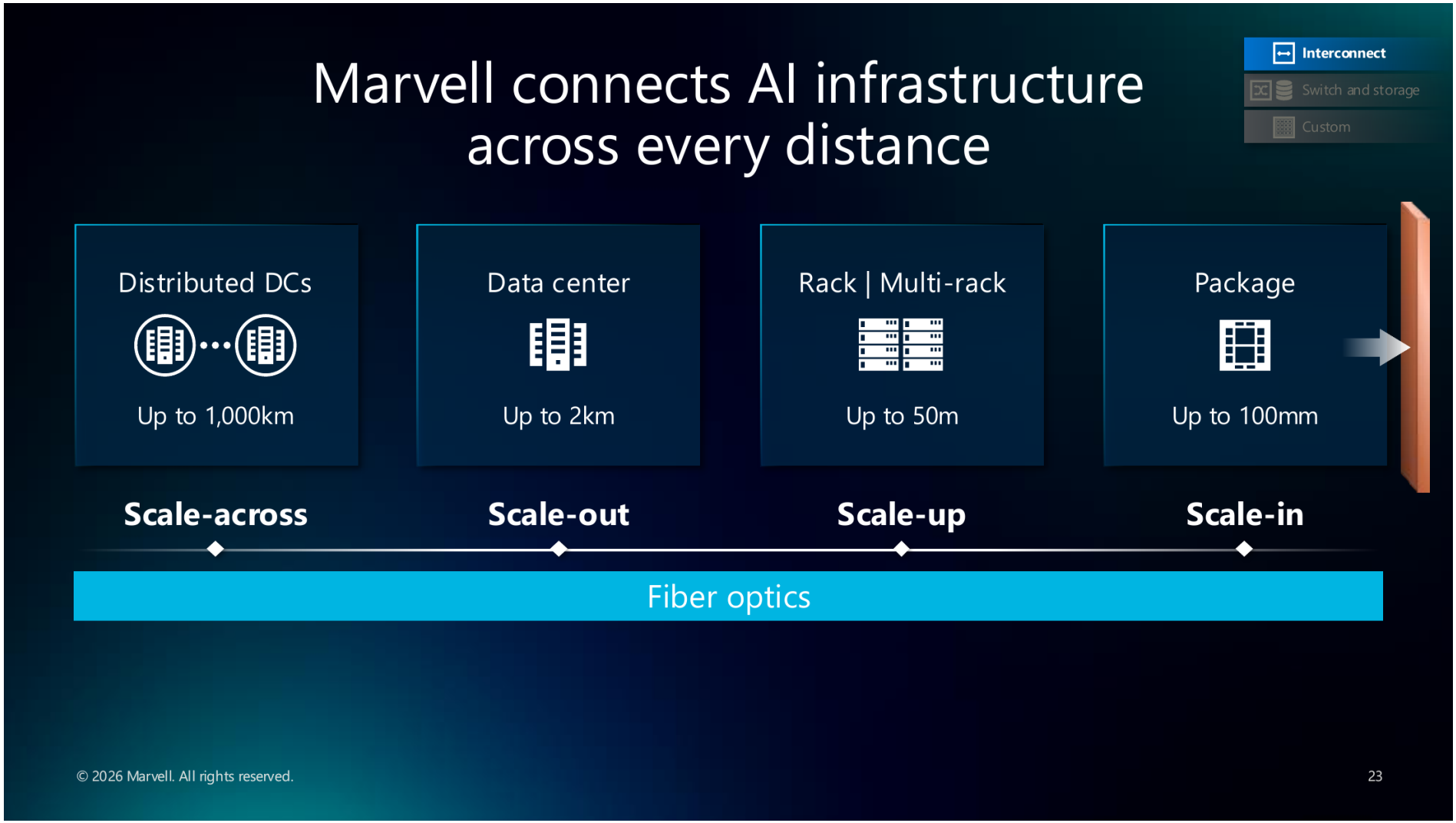
The company's CY2030 TAM is USD400B: USD65B interconnect, USD85B switching/storage, USD235B custom and USD15B other. These components sum to the total. The larger USD1.6T accelerated-compute infrastructure opportunity is nested inside the roughly USD3T capex envelope and cannot be added to Marvell's sales. The earlier USD94B market benchmark had a different date and scope. [ID 43-49]

The three-phase lens separates established sockets, qualified new attach/scale-up content and later architectural expansion. Its thematic weights are not applied again to already probability-weighted revenue.

Nine revenue vectors, with one invoice counted once

Modeled vector	Product and customer value	Cost and overlap boundary
Custom XPU and attach, including Google	Full custom processors and semi-custom NIC, memory, storage, infrastructure-management and inference-acceleration silicon. Reused Marvell IP can shorten development and improve attach economics.	Contains custom memory/storage sales and Google-covered programs. Customer-funded engineering and proprietary IP support OP, but advanced wafers, packaging and mix constrain GM. No second USD120B Google layer or separate SRAM royalty.
Merchant Ethernet and electrical connectivity	Teralynx switches, electrical connectivity and related merchant networking products. Larger radix and lower link latency can improve accelerator utilization.	Switch and AEC revenue remain distinct from optical engines. Fewer network tiers can reduce link counts per accelerator; do not assume every bandwidth doubling doubles device units.
External optical DSP	PAM4 and coherent DSPs sold externally for 800G, 1.6T and later generations. Link robustness, security and power drive differentiated variants.	Excludes DSPs already inside a Marvell full-module invoice. Flat Pipe integrated optics can eliminate a DSP rather than adding a DSP to every future optical link.
External broadband analog	Drivers and TIAs sold to independent optical suppliers across pluggable, NPO and CPO designs. Noise, linearity and output swing affect the entire link.	Excludes analog components included within Marvell's own complete-engine or module invoice. SiGe/InP process and test yield influence COGS; a foundry capacity agreement is an input, not another sale.
Coherent and DCI modules	Complete modules for campus, metro and longer-reach links, alongside the separate merchant-DSP route to market.	A module invoice includes internal components and their costs. External laser sourcing, photonic integration and final test affect margin. Industry module units are not company shipments.
Merchant CXL and memory fabric	Structera expansion, pooling and near-memory compute products improve use of existing memory capacity.	Excludes custom attach already in the custom vector. DDR4 reuse is not new memory-bit production and cannot substitute universally for HBM bandwidth.
Celestial Photonic Fabric	Optical scale-up connections and prospective memory-disaggregation products. Low latency and integrated optics can improve the economics of shared compute and memory.	Retains its distinct acquisition-related recognition and earnout treatment. Sampling an appliance does not add a second full revenue stream over existing Photonic Fabric assumptions.
Other scale-up optical products	External NPO/CPO light engines and other optical products outside the Celestial line. Marvell can sell integrated engines or components to module suppliers.	Residual after removing Celestial and established optics. Technology, form factor and end-market labels must not create multiple claims on one invoice. Launch cost and optical packaging yield can dilute early margins.
Communications and other	The remaining communications and non-data-center business.	Reconciles to company end-market totals. Security or infrastructure opportunities that are already included in another product invoice receive no duplicate addition.

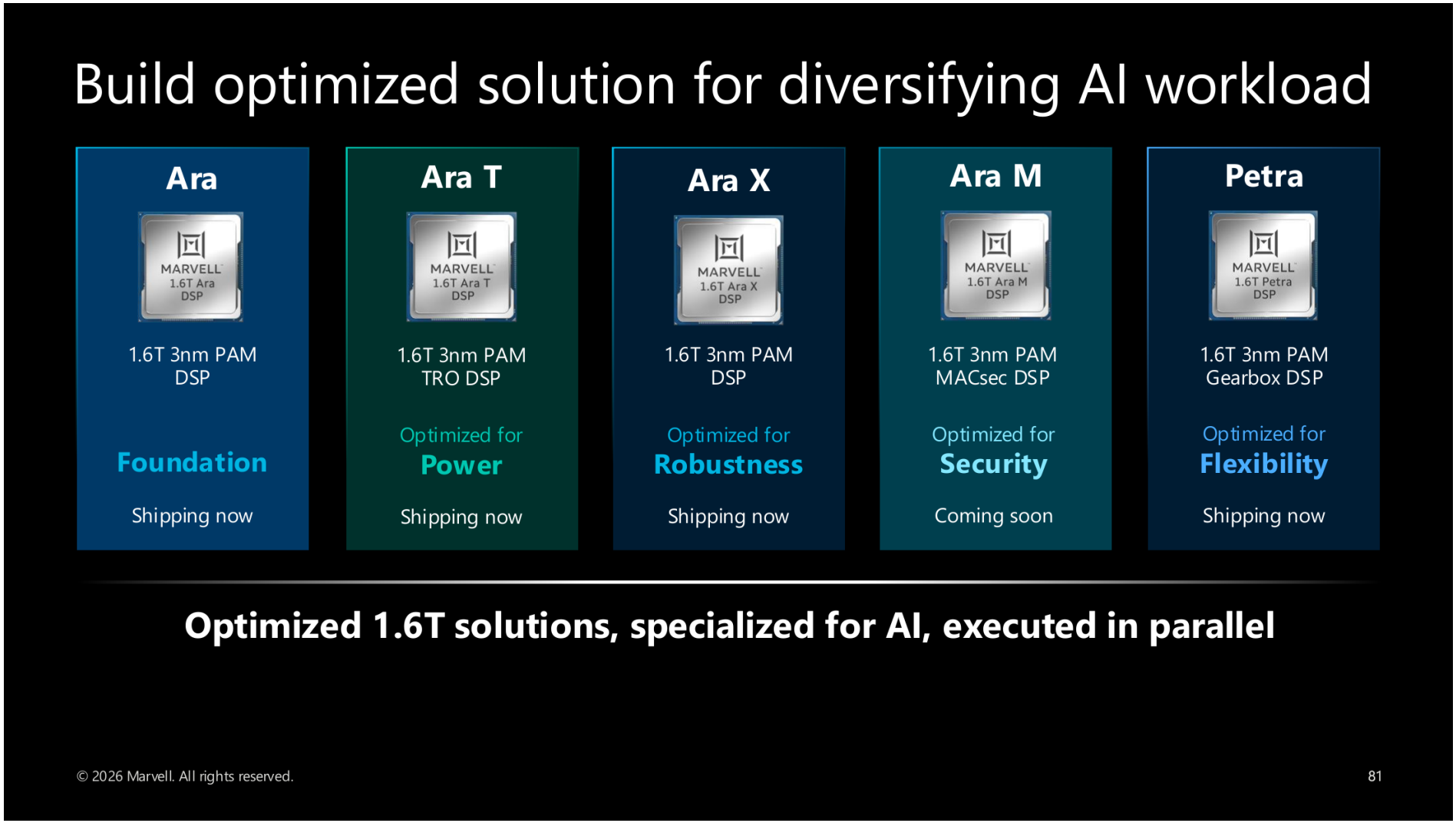
FQ2 FY2027, ended August 1, reported USD2,739.3M revenue and USD1,003.2M non-GAAP OP versus USD459.7M GAAP OP. Product-level allocations and margins remain estimates, not reported segment results. [M1]



Marvell Investor Day, October 6, 2026, PDF page 23. The connectivity map separates scale-across, scale-out, scale-up and scale-in. Reach labels describe the illustrated portfolio, not universal physical limits or additive company revenue pools.

Why the optical transition changes revenue and profit

Ara 1.6T is shipping in volume. At Investor Day, Ara, Ara T, Ara X and Petra were specifically marked shipping now; security-focused Ara M was coming soon. Variants optimize power, reliability, retiming, security and lane conversion. Their different maturity and use cases do not multiply the same module-demand pool. [T1; DECK p81]



Marvell Investor Day, PDF page 81. Ara, Ara T, Ara X and Petra are marked Shipping now; Ara M is Coming soon. Parallel variants serve different requirements and should not each be assigned the same total market volume.

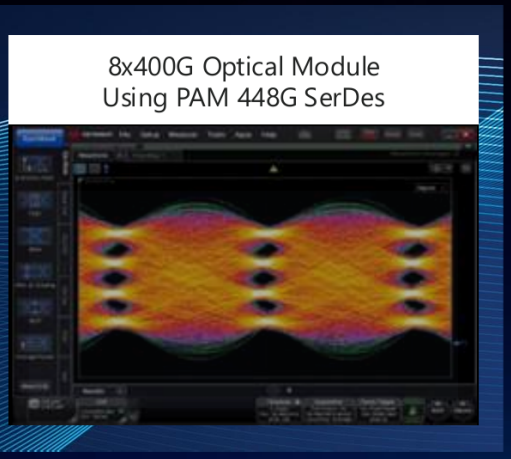
Generation changes can increase ASP as well as bandwidth. At 400G per lane, reach and link budget determine modulation; coherent-lite complements PAM4 for longer links. The 2nm 3.2T demonstration combines eight 400G PAM paths or two 1.6T coherent-lite paths. It is aggregate module bandwidth, not one 3.2T coherent lane or accepted-volume sales. The separate longer-reach roadmap places 1.6T ZR/ZR+ on 2nm/TFLN and future 3.2T on 14-angstrom technology with plasmonics for ZR/ZR+. No undisclosed commercial date is inferred. [ID 191-201; T2; DECK pp84,96]

Marvell industry-first 2nm live demos at ECOC

3.2T PAM4

Refreshing the definition of bandwidth

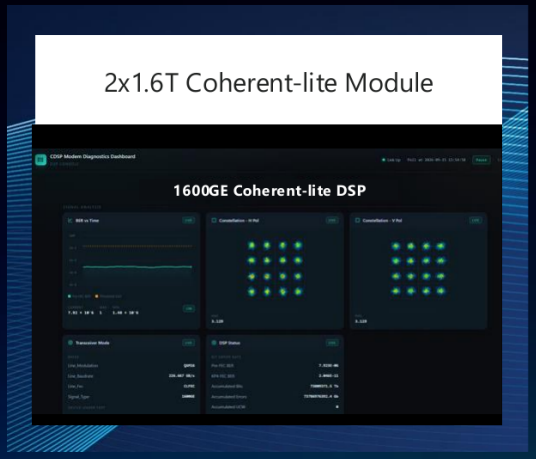
8x400G Optical Module
Using PAM 448G SerDes



3.2T Coherent-lite

Data center and campus reach, balanced power, and cost

2x1.6T Coherent-lite Module

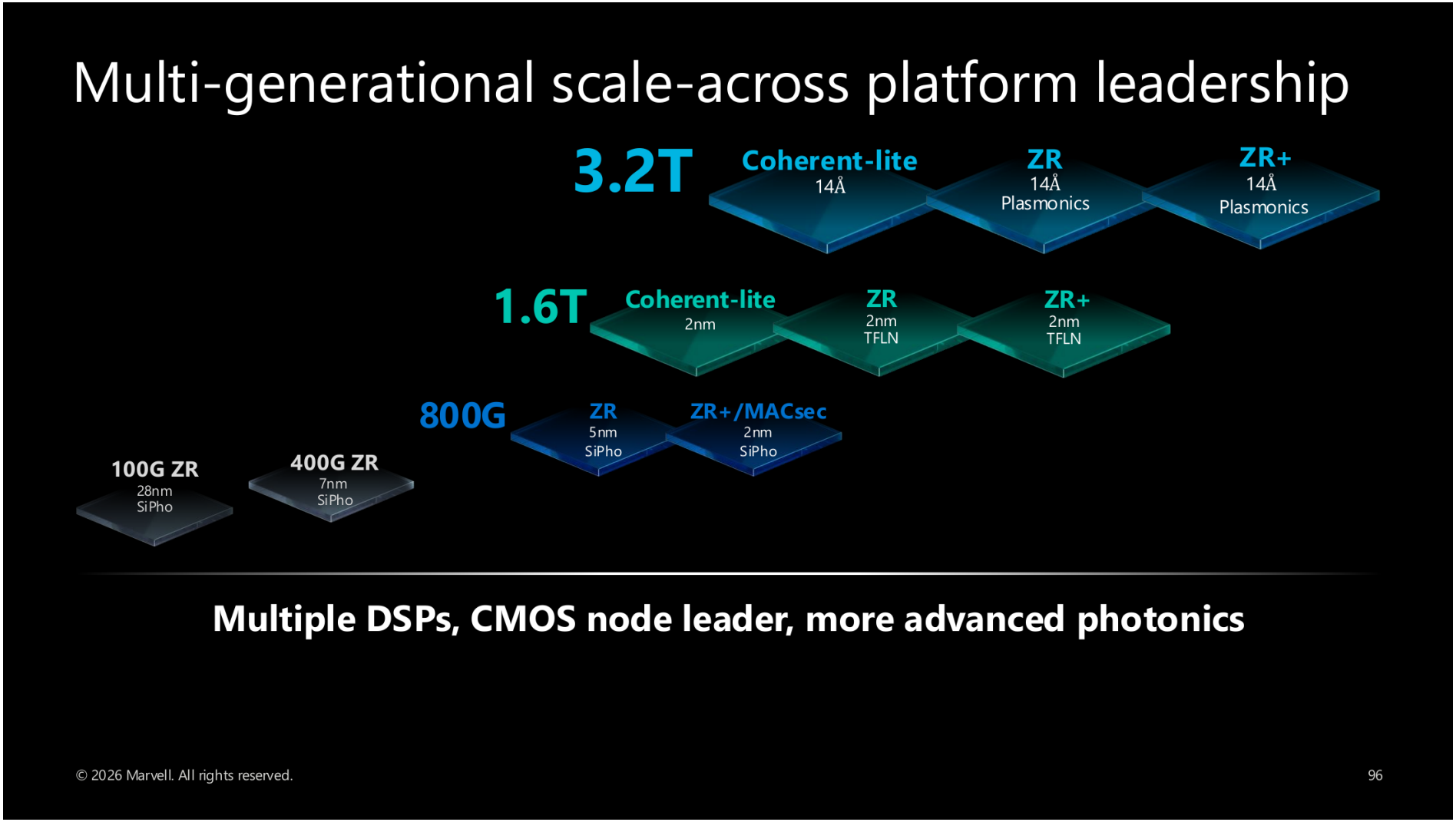


Path to 3.2T and beyond

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84

Marvell Investor Day, PDF page 84. The 2nm 3.2T demonstrations combine 8x400G PAM or 2x1.6T coherent-lite paths. This is aggregate module bandwidth and demonstration evidence, not a single 3.2T coherent lane or volume-shipment proof.

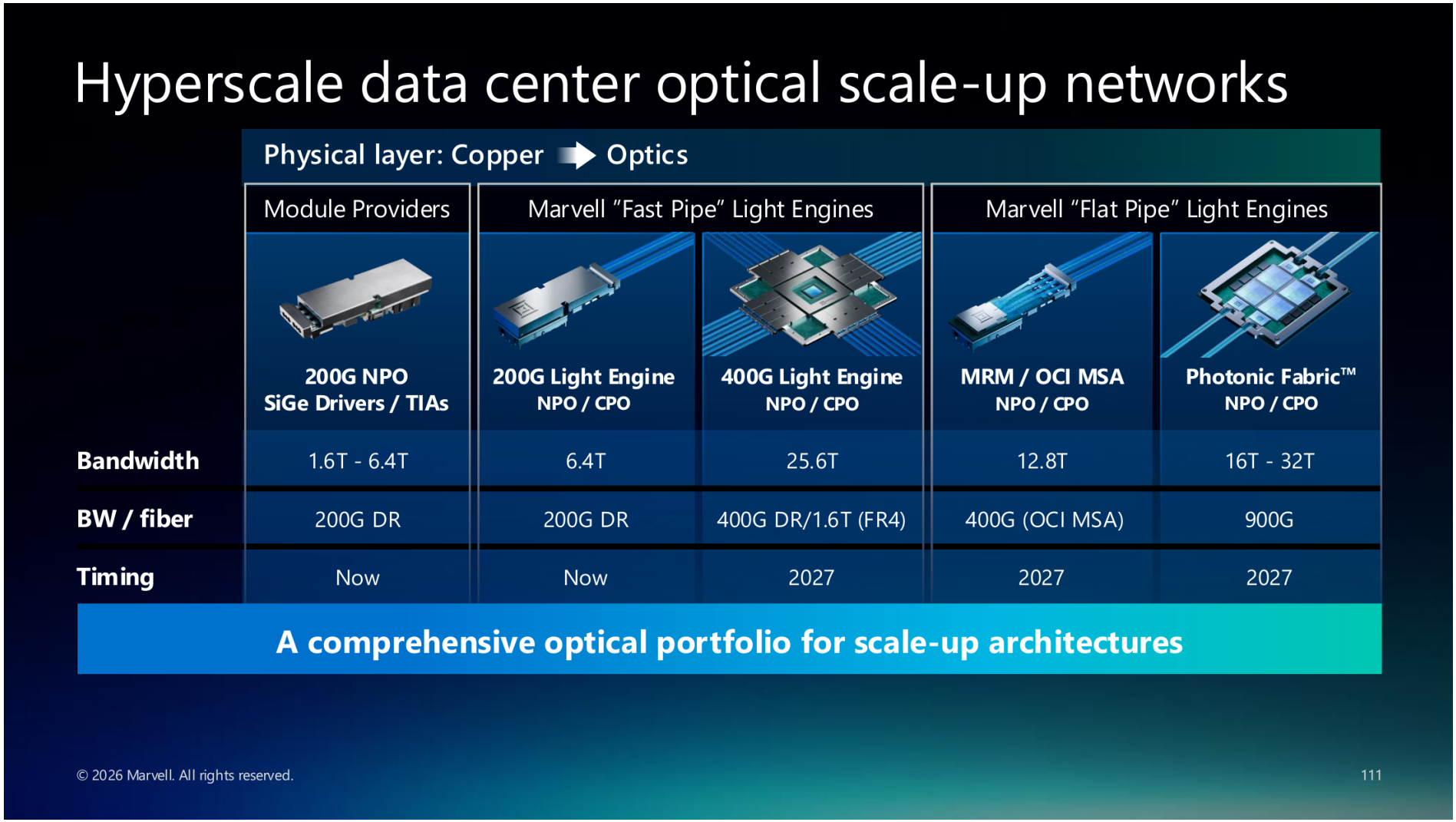


Marvell Investor Day, PDF page 96. Longer-reach roadmap: 1.6T on 2nm, including TFLN for ZR/ZR+, followed by 14-angstrom 3.2T and plasmonics for ZR/ZR+. Do not conflate this roadmap with the 2nm aggregate scale-out demonstration or infer undisclosed production dates.

Management sizes coherent-pluggable demand at just under one million units in CY2026 and over three million in CY2030, with its DCI business exceeding USD1B FY2028 from about USD500M FY2026. Complete-module and external-DSP routes coexist. The model's revenue anchors are not independently verified by an assumed module ASP: company shipments, net prices and yield remain material gaps reflected in probability. [ID 215-231]

Fast Pipe uses DSP/retiming options; Flat Pipe integrates control and analog equalization and removes the DSP. Photonic Fabric and OCI-based alternatives can use NPO or CPO formats. Replaceability and failure containment favor NPO; maximum integration can favor CPO. Management expects parallel ramps alongside copper.

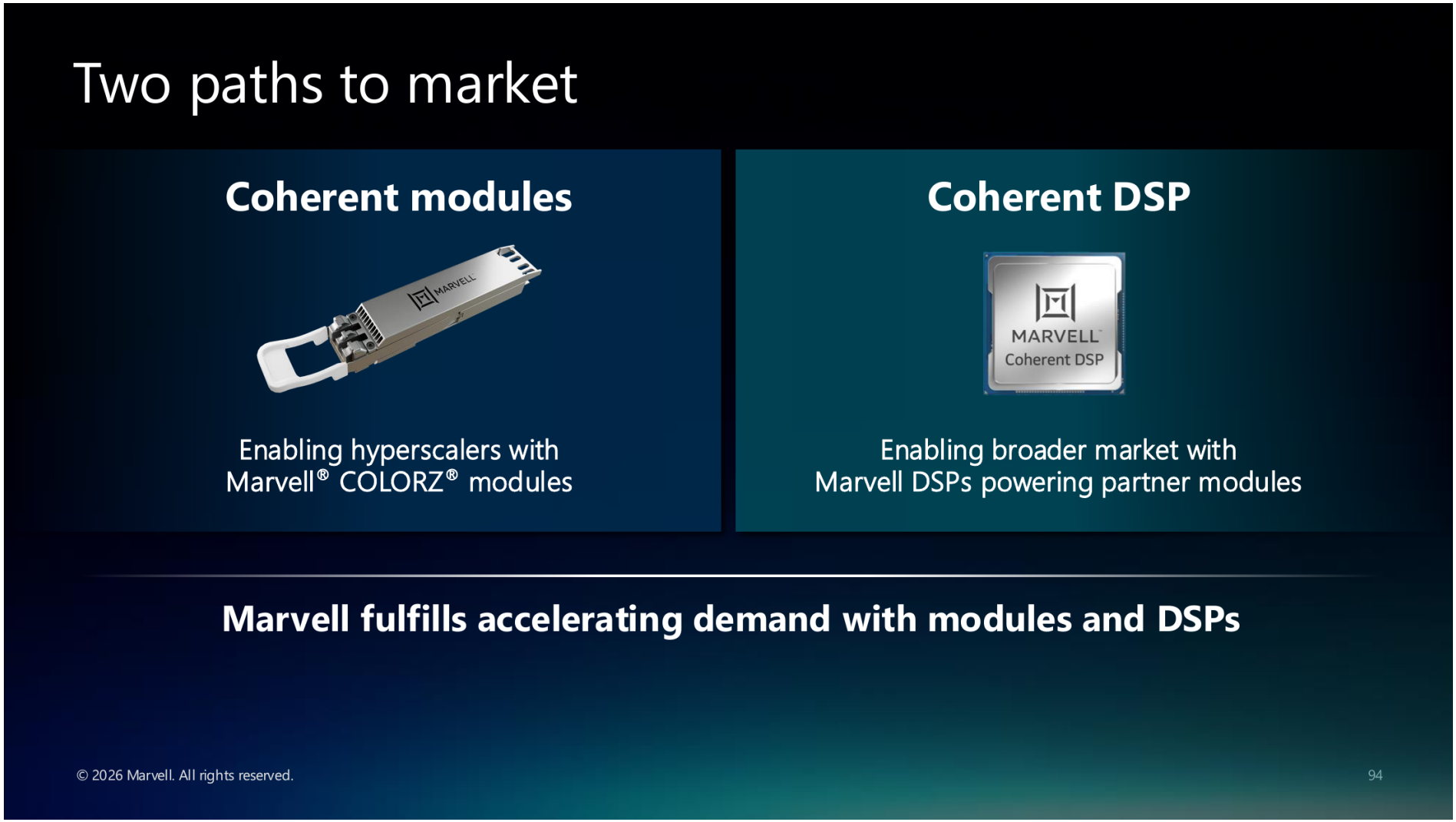
The roadmap's Now/2027 labels describe product timing, not accepted-volume revenue. [ID 271-285, 453-465; DECK p111]



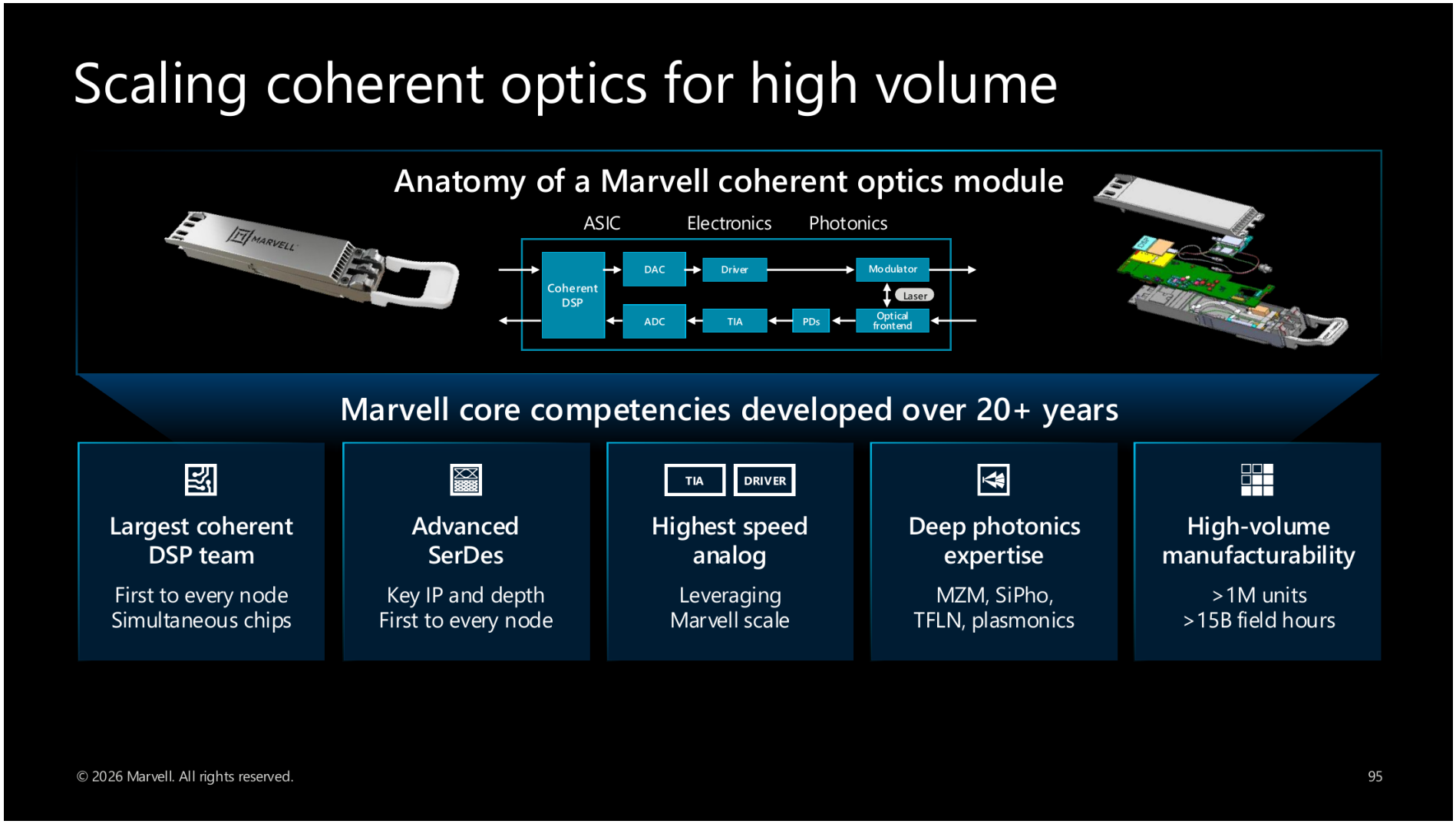
Marvell Investor Day, PDF page 111. Integrated-optics product roadmap spans light engines, OCI and Photonic Fabric. Now/2027 labels do not specify qualified volume, accepted sales or revenue-recognition dates.

Prospective optical content of thousands of dollars per accelerator is not a universal attach rate. Count both link ends consistently: a convention that already includes the switch end cannot be doubled again. Complete-engine revenue includes its electronics and costs. [ID 291-293]

No laser supplier is named. Co-development with leading manufacturers does not establish a new Lumentum, Coherent, Sivers or Aeva award. Supplier-specific capture needs its own evidence and earnings bridge. Complete modules and external DSP sales are distinct routes to market; cumulative field hours and installed units do not establish annual shipments. [ID 221; DECK pp94-95]



Marvell Investor Day, PDF page 94. Two routes to market: Marvell COLORZ modules and merchant DSPs inside partner modules. Internal components in a complete Marvell product do not create a second external invoice.



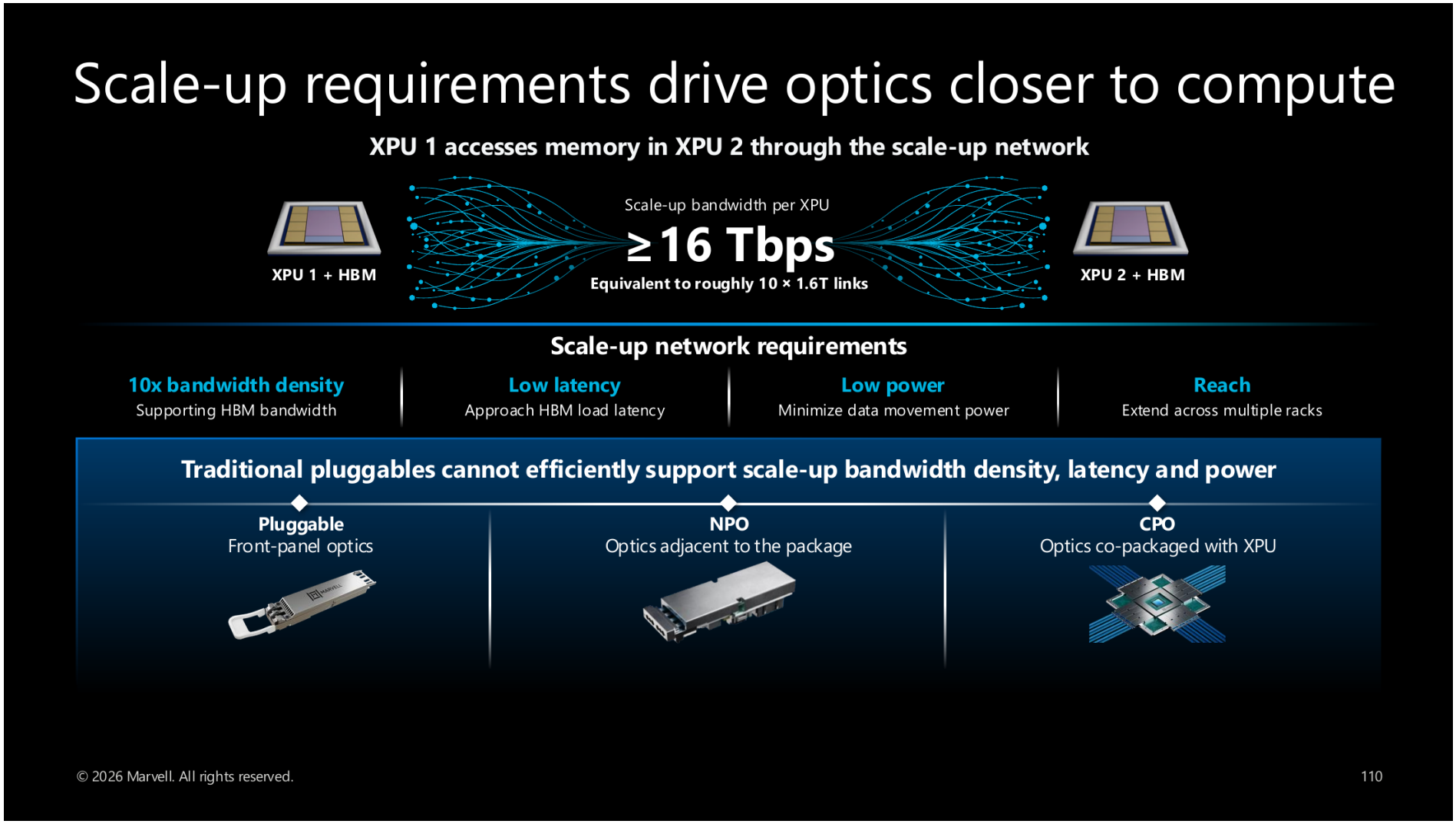
Marvell Investor Day, PDF page 95. A coherent module includes DSP, analog and optical components. Cumulative units and field experience are not annual shipments; the diagram does not identify an external laser supplier.

Switching, memory and custom attach

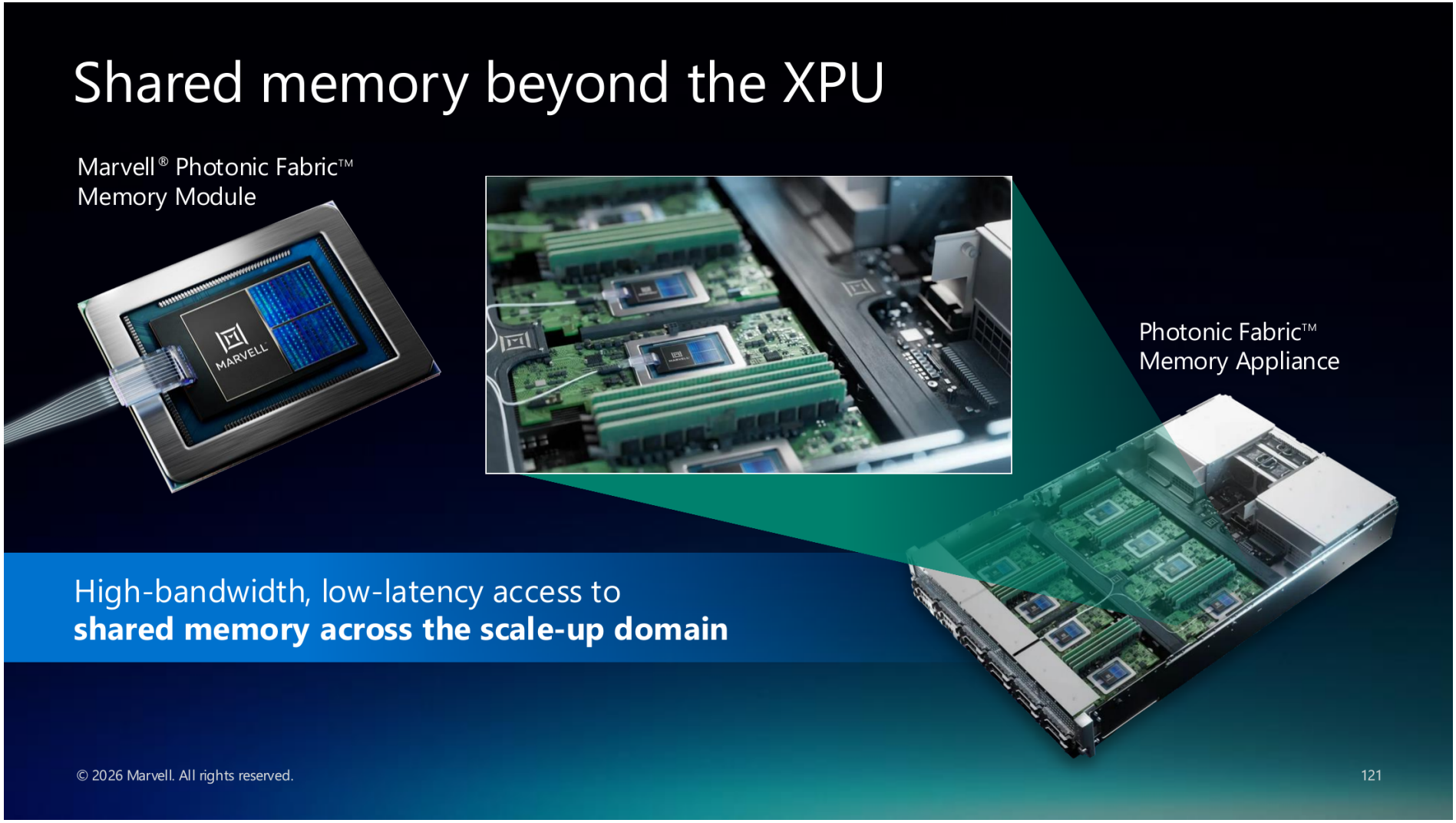
T100 is monolithic 102.4Tbps silicon with 512 200G lanes. Its case is lower power/latency and flatter topologies. Broadcom's Tomahawk 6 entered volume production in March 2026: monolithic implementation is the narrower Marvell distinction, not the absence of competing 100T products. Relative performance requires matched configurations. [T3; T4]

51.2T drives near-term switching growth. U115 UALink is expected to release/deploy in 2027; larger UALink and Ethernet/ESUN products remain roadmap items. NVLink Fusion adds choice, not simultaneous revenue from every protocol on every accelerator. [ID 239-247]

The Photonic Fabric memory module is in bring-up, with multi-customer sampling expected in 2027 and 8-16 modules per potential appliance. Independent memory scaling could avoid buying processors merely for their memory. Marvell's August technical description specifies up to eight DDR5 DIMMs and 72GB HBM cache per module, with 7.2Tbps optical-fabric bandwidth. This is separate from the DDR4-reuse controller path; see the supplier-effects discussion for the memory-content implications. [SO7] The scale-up slide specifies at least 16 Tbps per XPU, in bits per second; this is distinct from HBM bandwidth measured in TB/s. Approaching HBM latency is a requirement, not demonstrated equivalence. Integration, software, latency and acceptance determine economics; sampling adds no second revenue stream over the existing pool. [ID 287-289; DECK pp110,121]

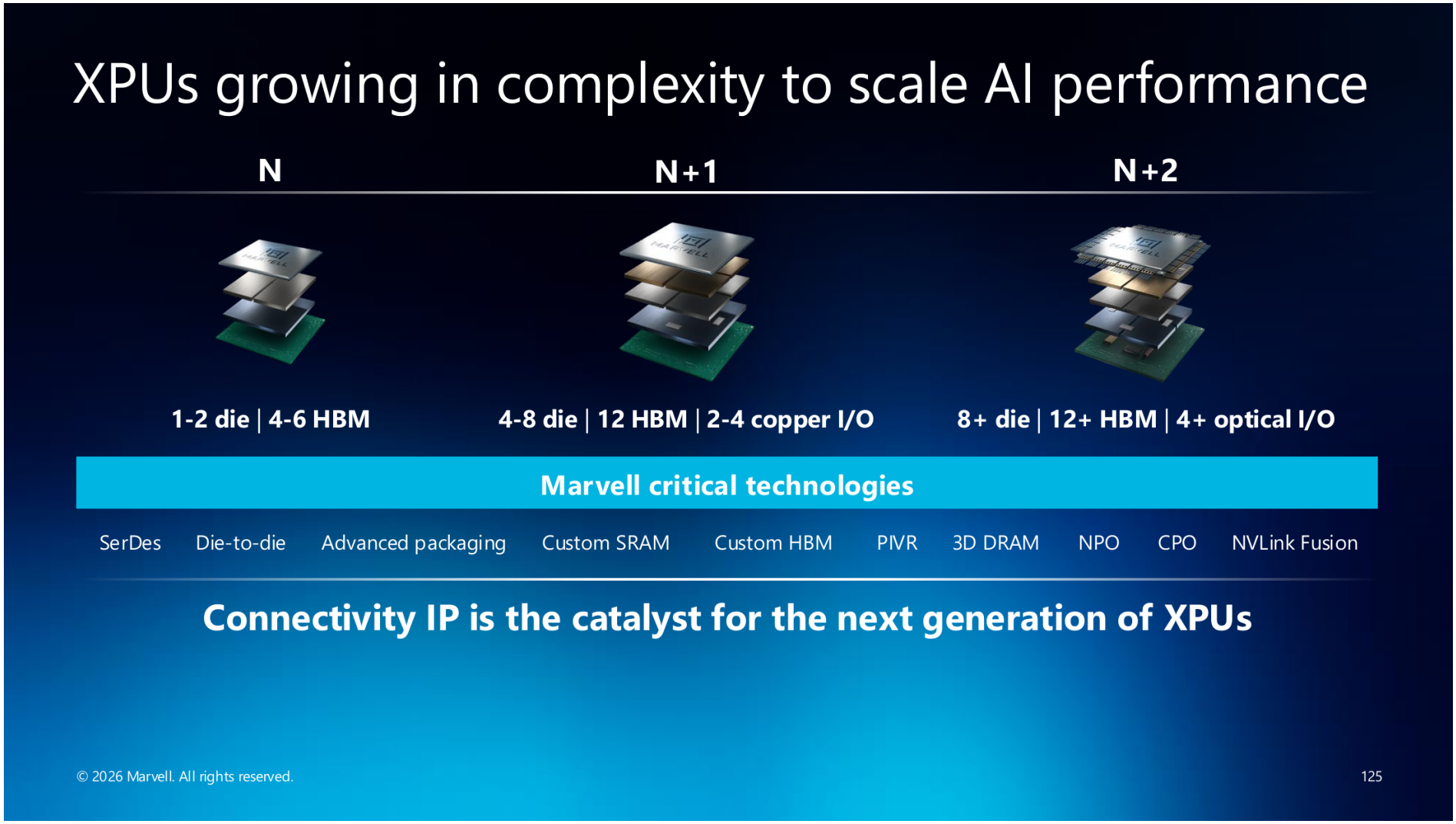


Marvell Investor Day, PDF page 110. The scale-up requirement is at least 16 Tbps per XPU, in bits per second. The latency wording is a design requirement, not proof of local-HBM-equivalent bandwidth or measured latency.



Marvell Investor Day, PDF page 121. Shared-memory appliance architecture explains independent memory scaling. Bring-up and expected 2027 sampling do not establish commercial volume, qualified capacity or net selling prices.

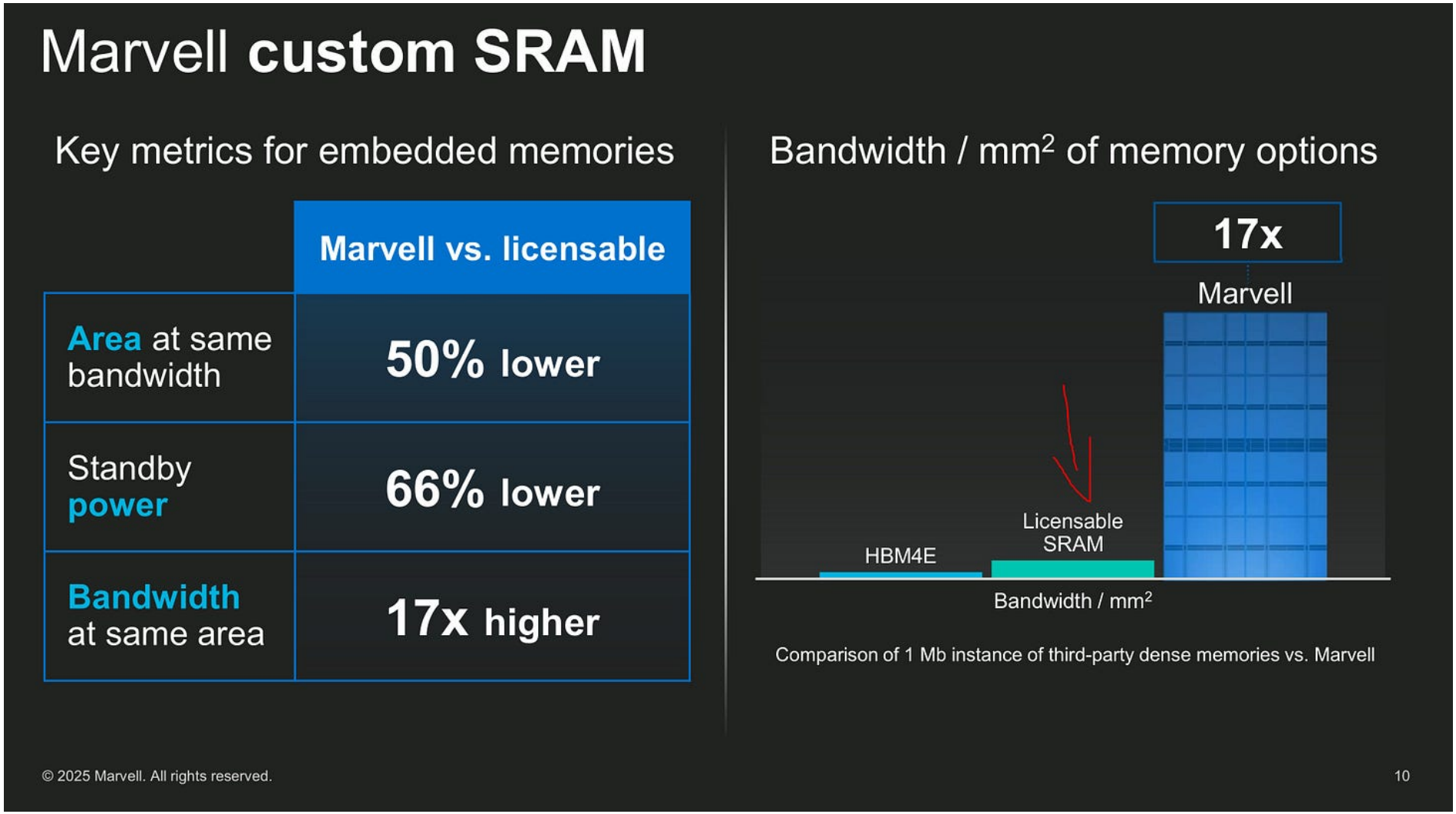
New attach categories are Octeon-based infrastructure management and specialized inference acceleration, alongside established NIC/memory/storage IP. Illustrative one-to-two sockets at about USD1,000 each are content clues, not a shipment schedule. Sole-source status applies to a customer program, not every future architecture. All these categories sit inside custom endpoints. [ID 305-329, 375-379]



Marvell Investor Day, PDF page 125. Illustrative XPU evolution increases compute die, HBM and I/O complexity. These are architecture examples, not a calendar shipment schedule or a disclosed bill of materials for every customer.

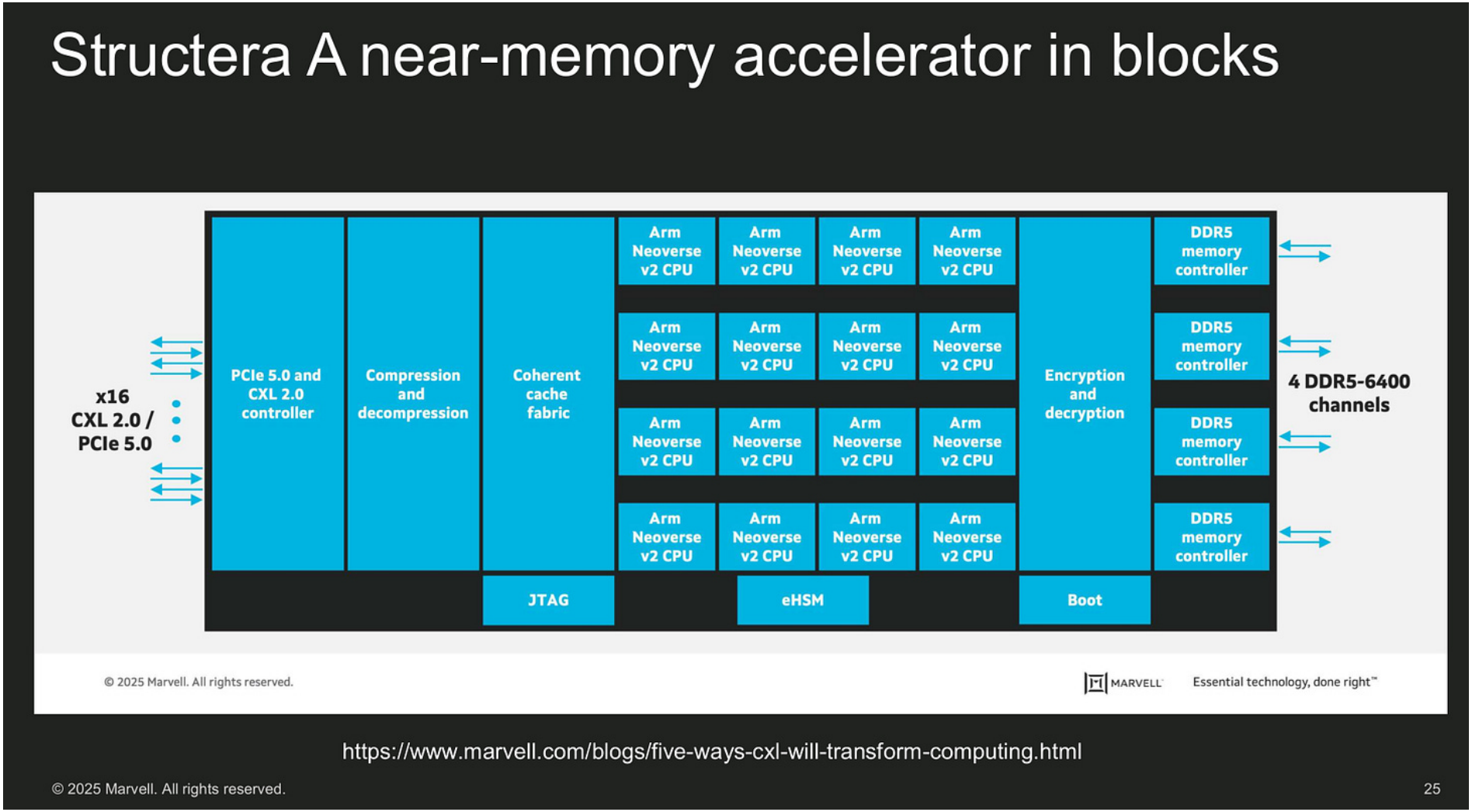
Custom SRAM and DDR4 reuse

The saved SRAM benchmark concerns memory-instance area, standby power and bandwidth per area versus licensable IP. It is not a whole-XPU comparison or proof of superiority over every proprietary competitor. The author's comparator-vendor attribution remains a guess. [T6]



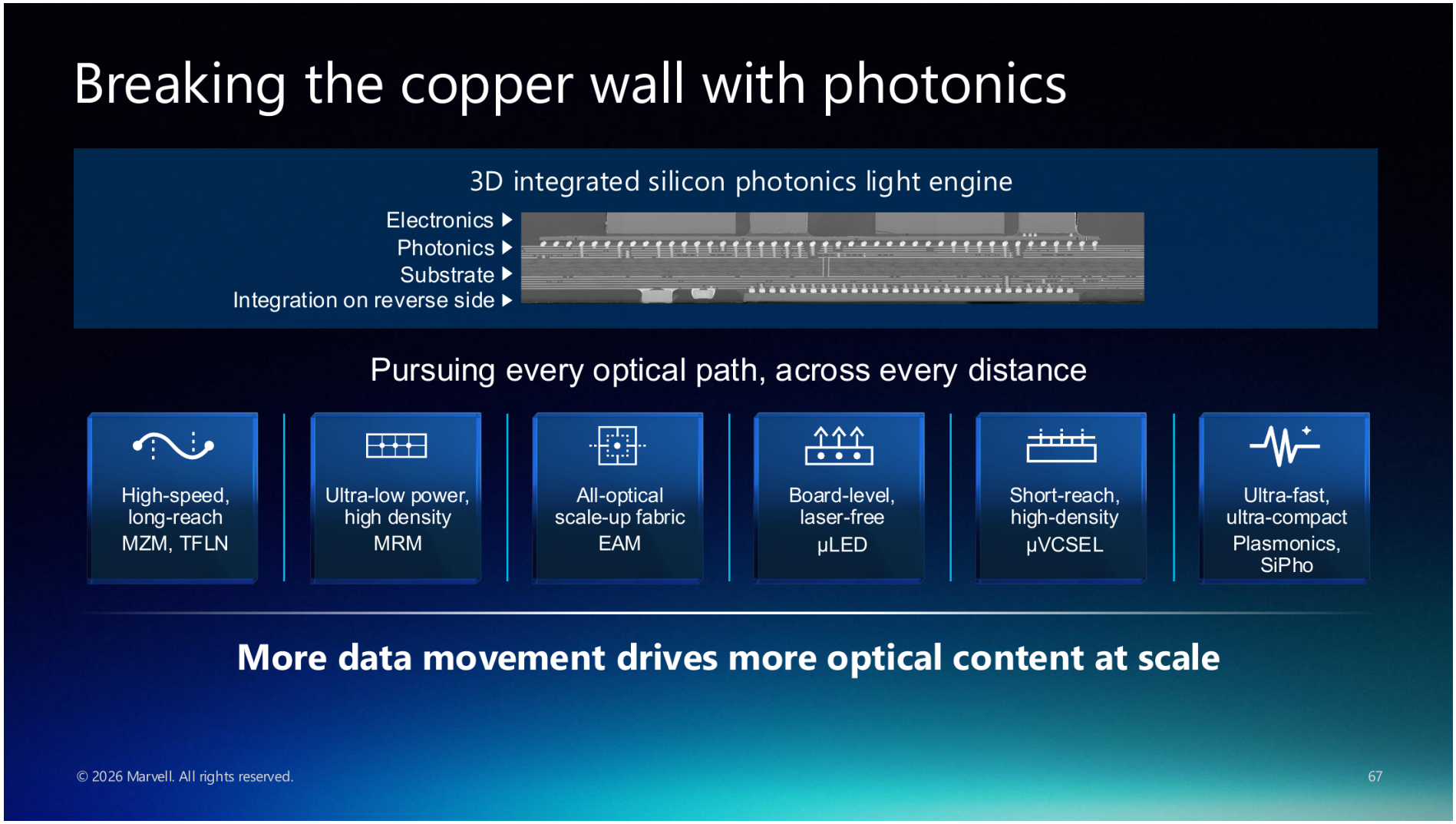
Saved Marvell SRAM comparison: benchmark scope is a 1Mb memory instance, not an entire AI processor.

Structera X expansion, A near-memory compute and S pooling differ. The DDR4 X2404 is not the pictured DDR5 A device. DIMM reuse requires compatible hosts, software and acceptable latency; compression is workload-dependent. Reuse improves existing capacity rather than adding HBM supply. [T7]



Structera A near-memory accelerator with DDR5: this diagram is not the DDR4 Structera X controller.

SRAM and reuse already support custom/merchant-memory pools. No second royalty, controller sale or high-margin pocket is added above the same blended GP allowance.



Marvell Investor Day, PDF page 67. Multiple optical technologies address different distances and integration needs. A technology label is not a named supplier award or a separate incremental revenue stream.

Scale-in: real development, later commercial timing

Micro-LED and micro-VCSEL links offer many parallel channels, but emitter, receiver, fiber, electronics and packaging integration govern useful performance. Marvell's disclosed development agreement and investment identify private Mojo Vision as a partner; they do not identify every micro-emitter supplier as a beneficiary. [T8]

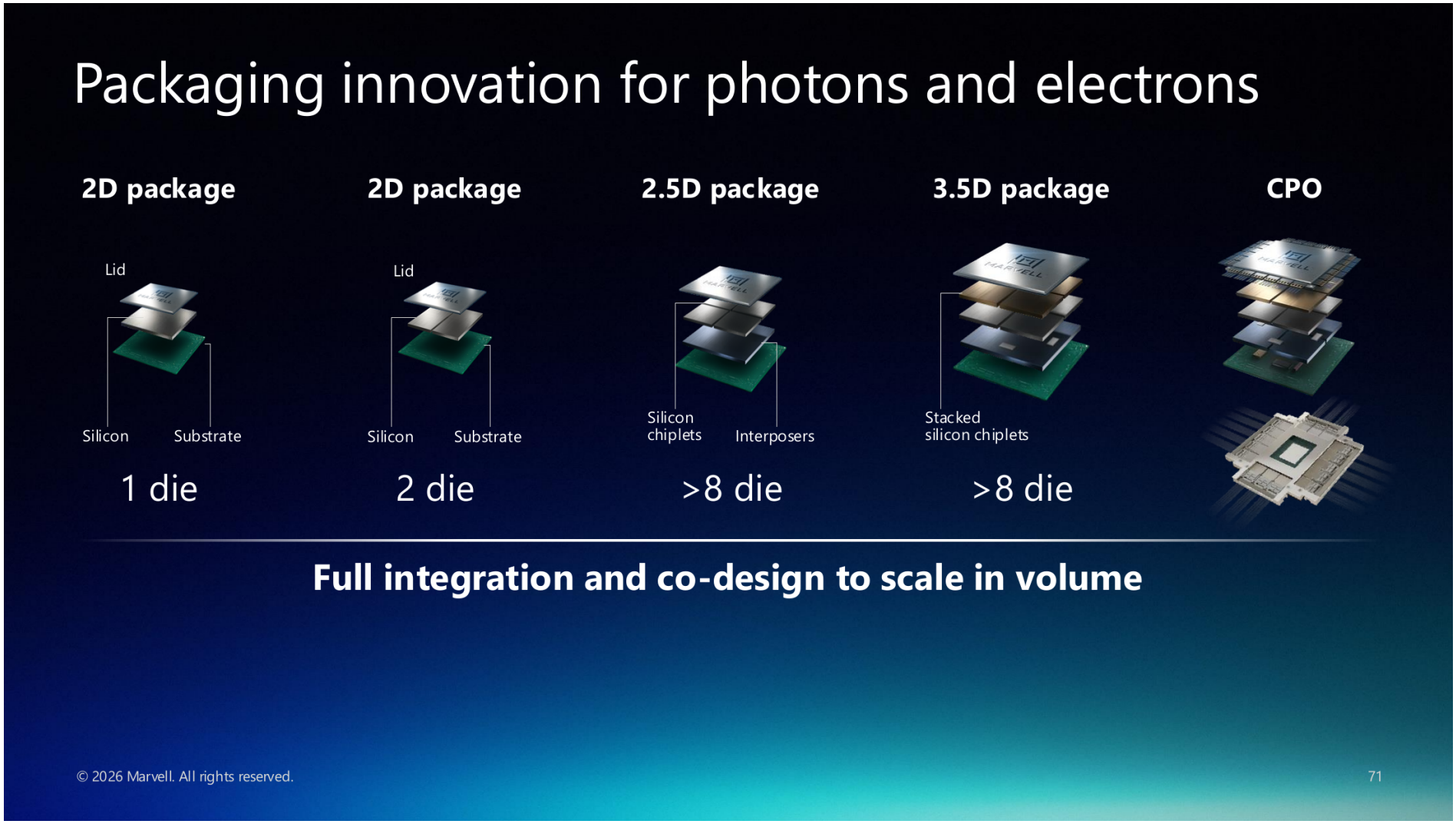
Scale-in is not material in management's FY2031 plan, although it could begin within the period. CY2029 timing came from an analyst's question. Development and qualification costs remain; no separate multi-billion line is added to that forecast. [ID 403, 505-509]

Capacity, costs and cash

Multi-year supplier forecasts and reservations span wafers, substrates, assembly and optics. Supply assurance supports the USD20B plan but is not a yield guarantee. Packages with 8-12 dies and sides approaching 150mm make qualified good-unit output more useful than nominal floor area. [ID 115-125, 137-165]

AT&S Kulim and GF SiGe agreements support product supply without adding another sales layer. Higher ASPs bring wafer, packaging and test costs. Capacity prepayments affect cash/working capital before inventory reaches COGS; do not expense them twice. [M4; M5]

Custom has lower GM but benefits from funded engineering and scale; attach can be richer because of proprietary IP. The CFO's long-range 56-59% GM, 44-46% OP margin and half-rate opex growth constrain the aggregate model, not every product or quarter. [ID 351-357, 487-491]

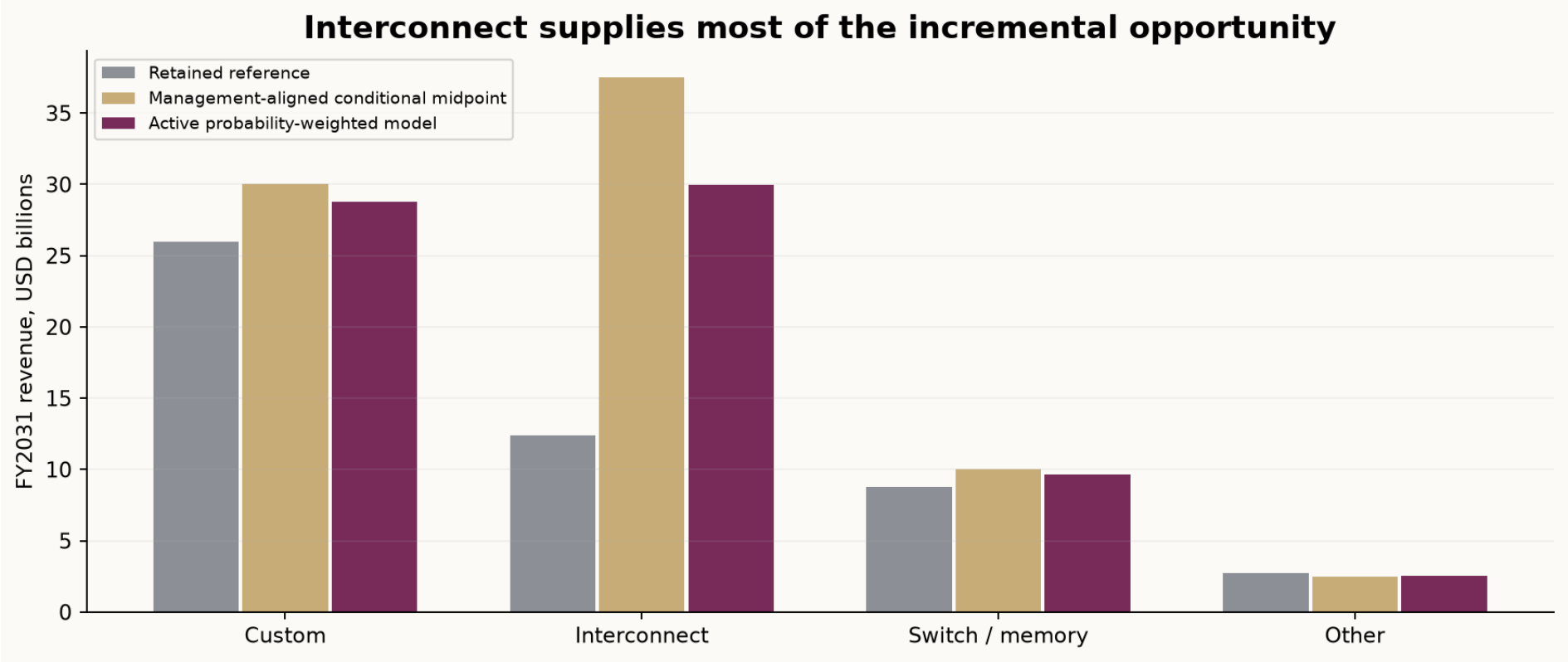


Marvell Investor Day, PDF page 71. Packaging and heterogeneous integration underpin the roadmap. Architectural capability does not quantify qualified output, yields, production cost or customer acceptance.

Probability, timing and the revenue-to-profit bridge

FY2028 is anchored at management's $20B company outlook. The 85% execution-confidence judgment is descriptive; it does not haircut that guidance. The estimated mix is $17.8B data center and $2.2B other, consistent with rounded "about $18B" data-center language and a low-growth other business. These product allocations are analytical estimates. FY2027 remains $12B, including the reported first half and the $3.15B FQ3 guidance midpoint.

For FY2029-FY2031, the active value for each product's revenue and gross profit, and for common operating expense, equals the retained reference plus **70% × (conditional outcome - retained reference)**. This applies the judgment once to incremental economics. The reference already contains Google, Celestial, established optical growth, custom attach and supplier capacity support. The longer-range weight reflects named programs, current production and planned supply, offset by correlated hyperscaler forecasts, qualification, yield and long lead times. The conditional endpoint is bounded by management's $70-90B FY2031 range; $80B is the selected midpoint. It is not accepted backlog.



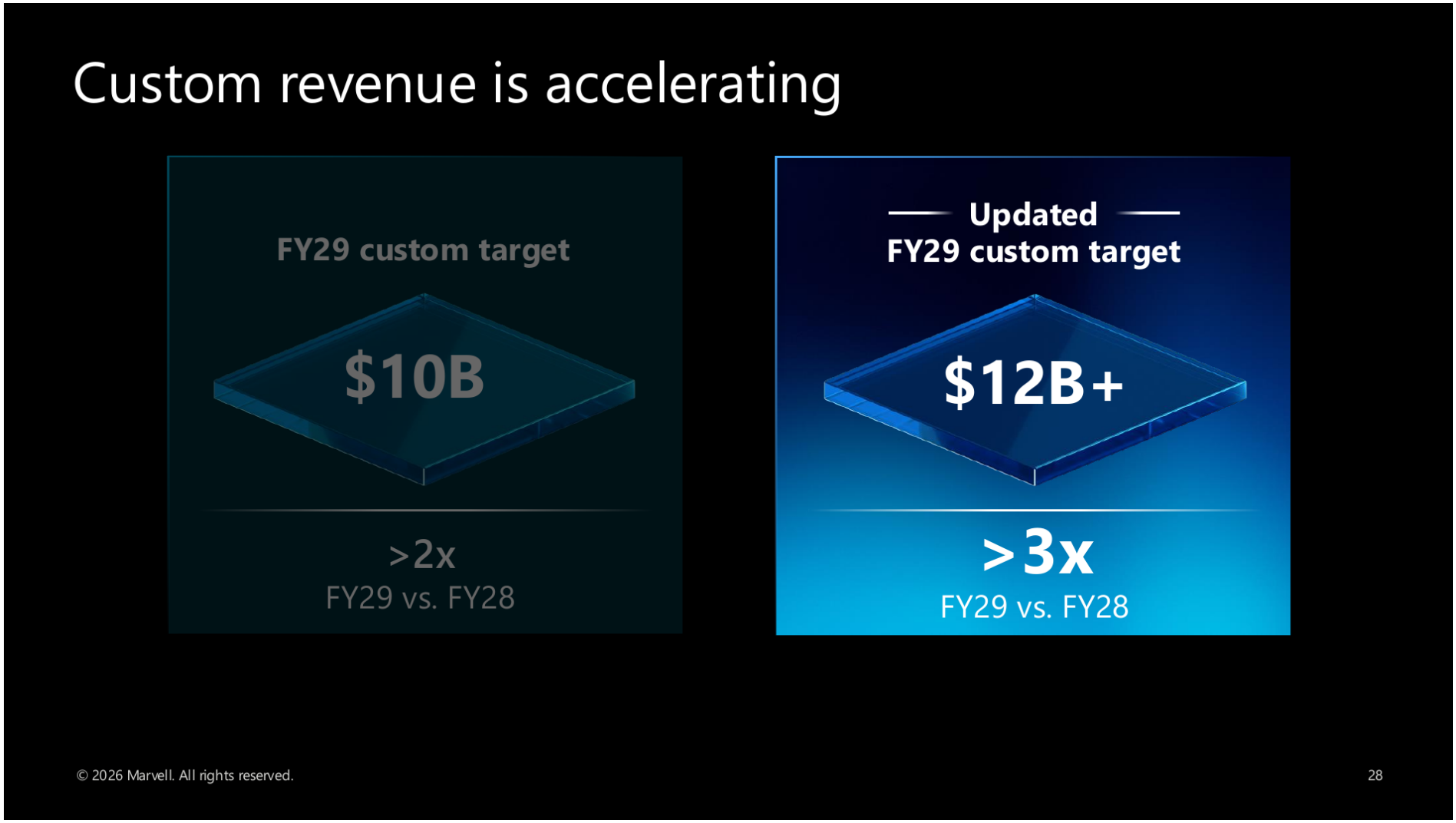
04 interconnect driver

FY2031 category	Reference, $B	Conditional, $B	70% increment, $B	Active, $B
Custom	26.000	30.000	2.800	28.800
Interconnect	12.400	37.500	17.570	29.970
Switch / memory	8.800	10.000	0.840	9.640
Other	2.700	2.500	-0.140	2.560

The 70% weight produces a $70.970B active FY2031 revenue diagnostic from the $49.900B retained reference. At the same weight, the $70B and $90B conditional endpoints imply $63.970B and $77.970B active revenue respectively; these are sensitivity bounds, not additional active cases. FY2031 is a fiscal-year diagnostic: the existing 18-quarter sheet ends at FQ3 FY2031 (calendar Q4 2030), so the missing FQ4 must not be silently treated as zero.

Established optics, other scale-up and switch/memory interpolate toward the selected FY2031 conditional allocation. Existing FY2029/FY2030 custom revenue of $14B/$20.5B remains in the model; the newly repeated >$12B FY2029 floor does not warrant adding another $2B. The original deck also states >3× FY2029 versus FY2028 custom growth (page 28); this is not merely a transcript ambiguity. The $12B+ floor and the ratio can coexist at a higher endpoint, but the current $7B/$14B model path does not satisfy the ratio. The deck does not disclose the FY2028 custom base needed to reconcile scope. No $21B-plus forecast is forced by multiplying the inherited $7B estimate. This same-event claim remains a specific model-to-source gap, rather than a second independent signal. The Google $120B opportunity, new attach categories and per-XPU content examples are all inside the custom envelope. Celestial keeps its separate recognition path.

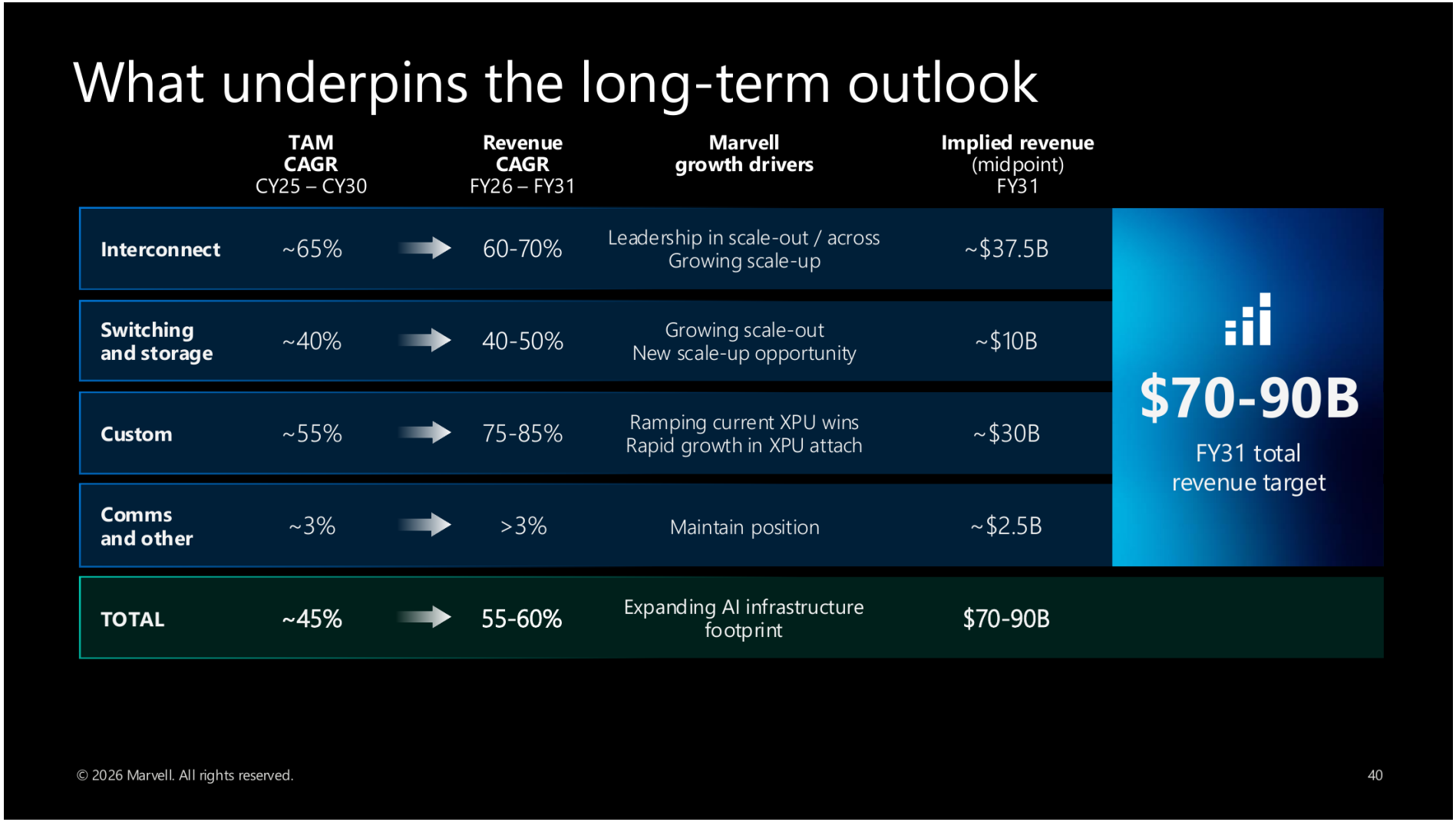
The merchant switch/storage grouping is mapped to the sheet’s existing electrical and memory/controller vectors as a proxy. Management does not disclose an exact product reconciliation. FY2028 DCI modules remain $1.320B because the >$1B company DCI statement is already covered; new established-optics allocation goes to external DSP and analog. No unverified module ASP is used to manufacture a company-unit forecast from global module demand. External component invoices exclude the components inside complete modules and engines.



Marvell Investor Day, PDF page 28. The original slide states FY2029 custom revenue of $12B+ and >3x FY2029 versus FY2028. The inherited $7B/$14B analytical path does not match that ratio; the deck does not disclose the FY2028 custom base here. The internal allocation gap is retained as an external clarification watch; the adjacent ratio analysis explains the zero additional consolidated increment.

Annual pre-policy model, USD B	FY2028	FY2029	FY2030	FY2031
Custom XPU / attach	7.000	14.000	20.500	28.800
Merchant electrical	2.900	3.749	5.029	6.573

Annual pre-policy model, USD B	FY2028	FY2029	FY2030	FY2031
External optical DSP	3.009	4.195	6.503	10.163
External analog	1.671	2.331	3.613	5.646
DCI modules	1.320	1.953	2.866	4.238
Merchant memory	1.000	1.480	2.174	3.067
Celestial	0.175	0.800	1.500	2.300
Other scale-up optics	0.725	1.249	3.016	7.622
Communications / other	2.200	2.360	2.460	2.560
Total revenue	20.000	32.118	47.661	70.970



Marvell Investor Day, PDF page 40. FY2031 company midpoint: $37.5B interconnect + $10B switching/storage + $30B custom + $2.5B other = $80B. These conditional management goals differ from the $70.97B probability-weighted active model.

What the custom-growth ratio actually constrains

The supplied slide28 and the full Investor Day transcript both say that FY2029 custom revenue exceeds $12bn and is more than three times FY2028. This is one event repeated in two formats, not two independent confirmations. The ratio's stated denominator is FY2028 custom revenue. No public issuer clarification located in this review supplies its exact dollar value or a reconciled program definition.

Let X be FY2028 and Y be FY2029 custom sales in billions. The claim is Y>12 and Y>3X. It does **not** establish X=4, Y=12, or a $21bn FY2029 target. The inherited X=7/Y=14 planning allocation is incompatible with that ratio. It remains visibly labeled an internal allocation gap, not a company forecast. The total-company model is anchored independently to FY2028 sales and consolidated costs; its custom gross margin is a balancing allocation rather than a disclosed product margin.

Illustrative ratio-consistent pair, USD bn	FY2028 custom	FY2029 custom	FY2029 / FY2028	FY2028 other DC within $18bn
Lower endpoint illustration	3.9	12.1	3.10×	14.1
Middle endpoint illustration	4.5	14.0	3.11×	13.5
Higher endpoint illustration	5.2	16.0	3.08×	12.8

These are bounded reconciliation examples, not three new sales forecasts. The $12.1-16bn interval and $3.9-5.2bn bases are analyst-selected illustration bounds, not a management upper limit. They show that a lower FY2028 starting point can satisfy the ratio without any increase in the adopted FY2029 $14bn endpoint. Multiplying the old $7bn assumption by three would instead manufacture an unsupported upgrade.

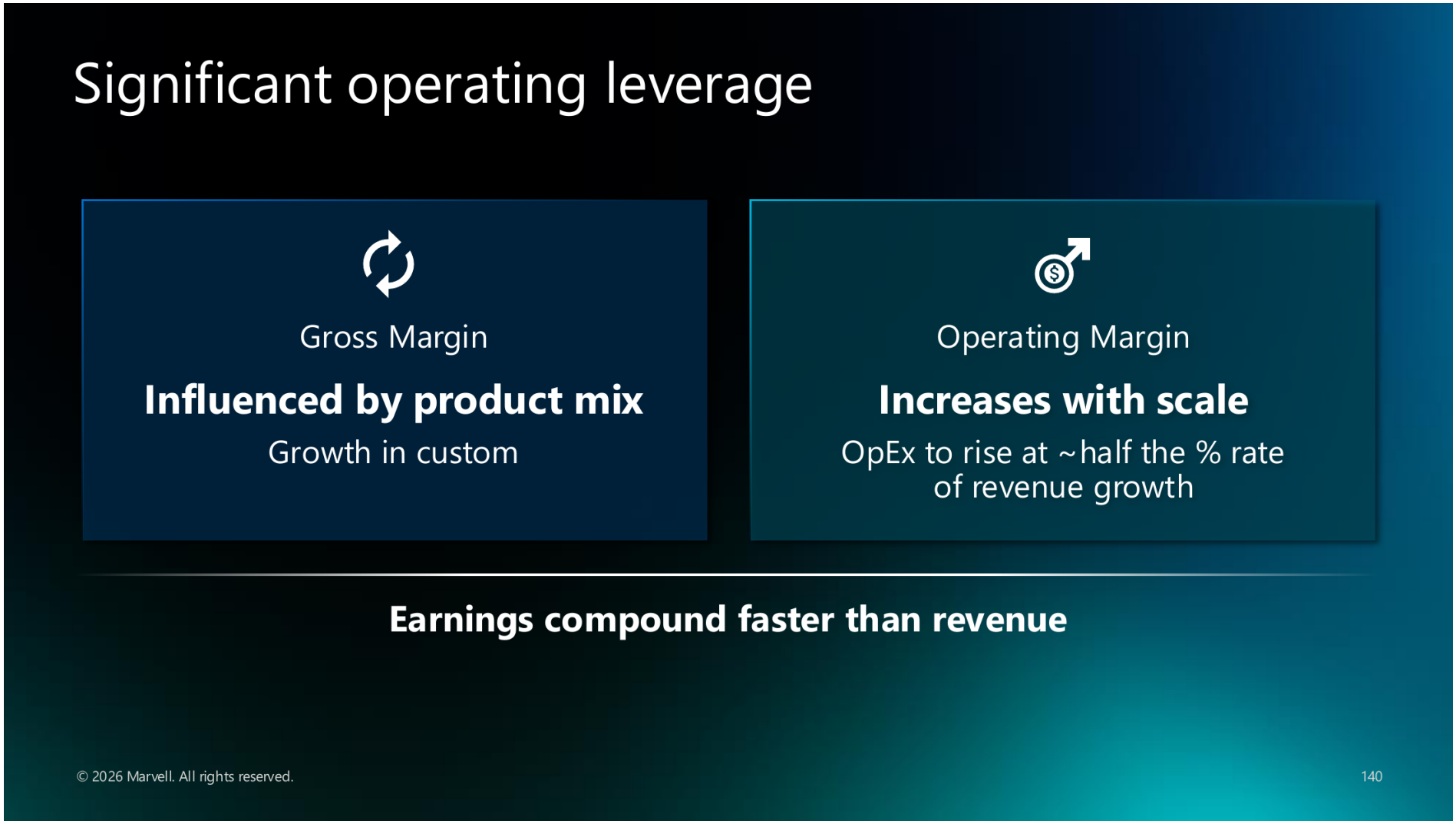
The source claim has a 70% commercial-realization weight within the existing Investor Day opportunity, and this review has 100% event overlap with that already-modeled evidence. Conditional **incremental** company sales are $0 in the allocation-only reconciliation: an FY2028 reduction of $2.5bn in custom, for example, is offset by $2.5bn in the remaining datacenter envelope. Conditional incremental consolidated GP, opex and OP are therefore $0 while the retained 58% consolidated GM and $3.4bn opex anchors govern. That is the reason for zero additional ceiling impact; it is not a zero probability assigned to management’s growth statement.

The product mix still matters economically. If $2.5bn moves from the current balancing custom margin of50.30% to an illustrative65% merchant margin **without offsetting costs**, GP and OP would rise by $367.5m. That is a sensitivity, not an active upgrade: the FY2028 cost reconciliation already fixes consolidated GP at $11.6bn, and no evidence establishes65% net profitability for the incremental optics/packaging mix. Customer funding, outsourced assembly, yields and launch R&D can absorb the apparent difference. Applying both that sensitivity and the current consolidated margin would count the same benefit twice.

The active Q4 2027 ceiling remains $227.7064 at20× forward-four-quarter non-GAAP OP. Analytical Base85 and protected Excel Base 85 are unchanged. The research check is complete; the external watch item is an issuer-supplied FY2028 custom base and program scope, followed by accepted custom and interconnect revenue. When that evidence arrives, replace the allocation and its associated product costs together. No correction to the source year or ratio has been invented.

Explicit costs and operating leverage

FY2028 revenue growth is 66.67% on $12B; applying the CFO’s approximate half-rate expense growth to $2.550B gives $3.400B opex. The retained 58% consolidated GM anchor gives $11.600B GP and $8.200B OP. The implied 50.30% custom GM is a balancing allocation, not a disclosed decline in custom product profitability. The annual cost reconciliation governs; product margins remain estimates.



Marvell Investor Day, PDF page 140. The CFO describes approximate half-rate operating-expense growth relative to revenue growth. This is an aggregate operating-leverage framework, not a promise of fixed costs or every-quarter margin expansion.

For the long-range conditional case, unchanged raw product-margin proxies would overstate integrated-optics economics. An explicit packaging, yield and qualification COGS reserve is assigned to other scale-up optics. Added common growth investment sits above mechanical half-rate expense growth. This aligns the conditional FY2031 midpoint with 57.5% GM and 45% OP margin, inside the CFO's 56-59% and 44-46% ranges. Intermediate margin trajectories are analyst assumptions, not year-by-year guidance. Active product GP and common expense are blended separately, so neither revenue nor margin is probability-discounted twice.

The CFO also frames a 15% long-term tax rate, free-cash-flow margin above 36%, and average capital returns exceeding 50% of free cash flow. These are management goals, not a disclosed quarterly cash or share-repurchase schedule. Capacity deposits, inventory, acquisition payments and excluded stock compensation still matter. The goals are cash-conversion checks; they do not replace operating profit in the valuation or automatically reduce diluted shares. [ID 355-361]

Conditional cost bridge, USD B	FY2028	FY2029	FY2030	FY2031
Revenue	20.000	33.454	50.944	80.000
GP before incremental COGS reserve	11.600	19.571	30.133	47.856
Added optical packaging/yield COGS	0.000	0.168	0.712	1.856
Gross profit	11.600	19.403	29.420	46.000
Mechanical half-rate opex	3.400	4.544	5.731	7.366
Additional growth expense	0.000	0.698	1.443	2.634
Total opex	3.400	5.241	7.175	10.000
Operating profit	8.200	14.162	22.246	36.000

Active annual economics, USD B	FY2028	FY2029	FY2030	FY2031
Revenue	20.000	32.118	47.661	70.970
COGS	8.400	13.558	20.200	30.190
Gross profit	11.600	18.560	27.461	40.780
Operating expenses	3.400	4.917	6.507	8.669
Operating profit	8.200	13.643	20.953	32.111
Gross margin	58.00%	57.79%	57.62%	57.46%
Operating margin	41.00%	42.48%	43.96%	45.25%

Existing pricing hypothesis: once, on eligible invoices

The existing FCC optics scenario retains a 40% probability, 50% qualifying exposure, 50% new/repriceable orders and a conditional 5% price premium. Incremental COGS and opex are each 5% of that price revenue. Phase intensity is zero before Q3 2027, then 25%, 50%, 75% and 100% from Q2 2028. Only external DSP and analog sales qualify; modules, integrated engines, custom and supplier revenue do not.

The October 6 outlook is not independent confirmation of a policy premium. Its product-generation growth defines the new eligible invoice pool; the separately contingent like-for-like premium is not disclosed as included in guidance. Thus the existing small policy layer is recalculated on that pool once. If contracts or the

eventual policy show the premium is already in quoted guidance, remove the overlapping portion. The shared 40% probability is unchanged across portfolio names; the exposures are correlated.

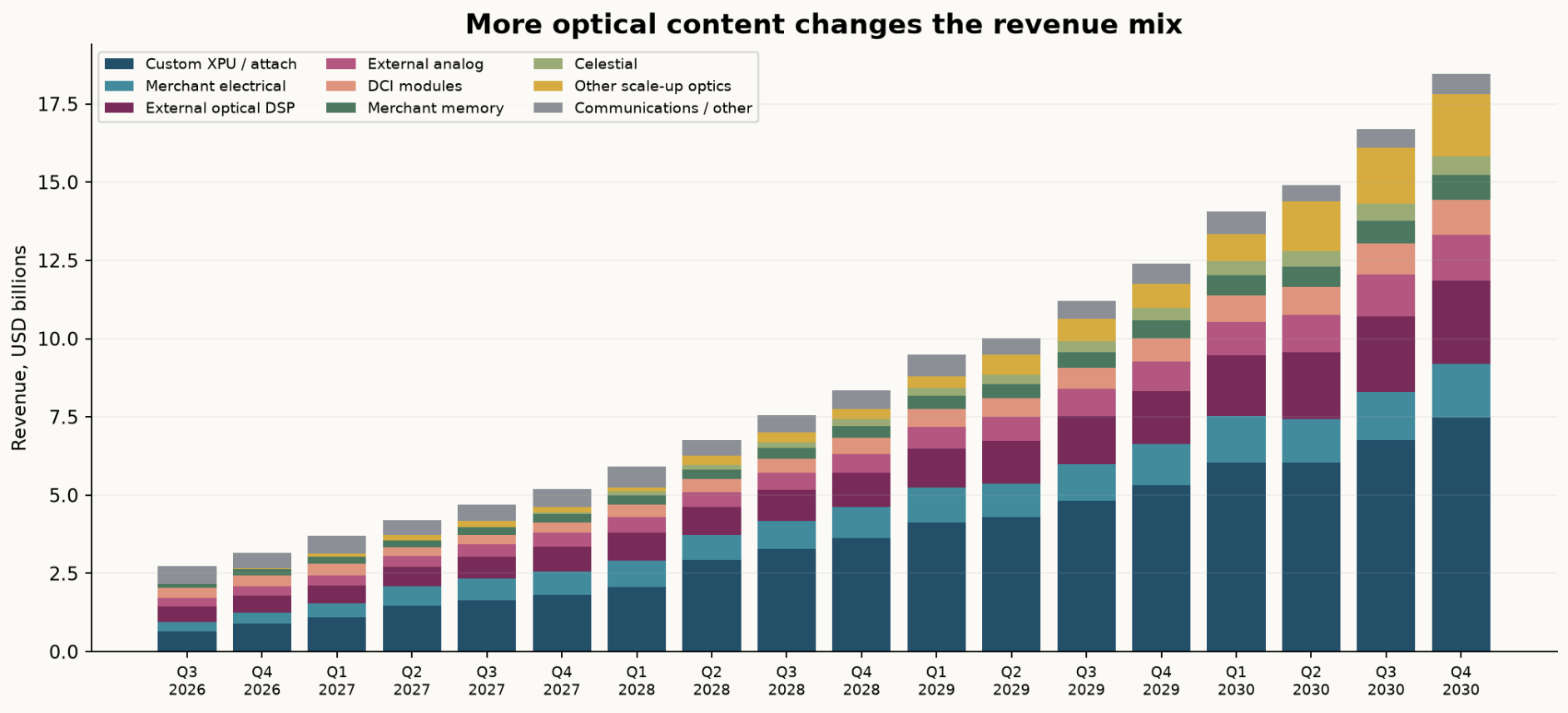
Quarter	Price revenue, $M	Added COGS, $M	Added opex, $M	Added OP, $M
Q3 2026	0.000	0.000	0.000	0.000
Q4 2026	0.000	0.000	0.000	0.000
Q1 2027	0.000	0.000	0.000	0.000
Q2 2027	0.000	0.000	0.000	0.000
Q3 2027	1.375	0.069	0.069	1.237
Q4 2027	3.042	0.152	0.152	2.738
Q1 2028	5.177	0.259	0.259	4.660
Q2 2028	6.853	0.343	0.343	6.167
Q3 2028	7.668	0.383	0.383	6.902
Q4 2028	8.484	0.424	0.424	7.636
Q1 2029	9.626	0.481	0.481	8.664
Q2 2029	10.621	0.531	0.531	9.559
Q3 2029	11.885	0.594	0.594	10.697
Q4 2029	13.150	0.657	0.657	11.835
Q1 2030	14.920	0.746	0.746	13.428
Q2 2030	16.600	0.830	0.830	14.940
Q3 2030	18.576	0.929	0.929	16.719
Q4 2030	20.552	1.028	1.028	18.497

What would change the weight

Signal	Numerical treatment	Next evidence
FY2031 interconnect / custom execution	70% joint incremental weight; bounded $70-90B conditional total	Increase with accepted production and customer allocations; reduce for delay, cancellations, yield loss or architecture displacement.
FY2028 company plan	100% guidance anchor; 85% execution confidence	A documented guidance cut or thesis break, not absent slides, would justify reducing the anchor.
CXL / DDR4 reuse	70% useful deployment judgment; 100% overlap with merchant-memory pool	Product mix and accepted volumes must exceed the current pool to create another increment.
DDR4 high-margin pocket	30% probability of 75-85% GM on 10-20% of FY2028 merchant-memory sales	The remainder's 57.5-61.7% GM fits the existing 63% blend; zero extra GP avoids counting it twice.
Custom SRAM	75% useful IP-advantage judgment, included within custom	No separate royalty invoice; benchmark scope and qualified design wins matter.
Scale-in optical I/O	50% judgment of some qualified deployment by FY2031; $0-0.4B sales diagnostic, chosen as at most 0.5% of the $80B midpoint, not company guidance	At assumed 30-50% GP margin and $0.1-0.2B allocation of existing R&D/launch spending, the conditional allocated OP range is -$0.2B to +$0.1B. These are contained within broad optical mix and common costs, not incremental invoices or costs. There is no supported volume timing inside Q4 2027-Q3 2028. Zero additive OP reflects timing and analytical overlap, not a requirement for official confirmation.
AT&S / GF availability	85% supply-execution judgments already embedded in product programs	Qualified good units, yields, allocated volume and cost determine an increment; reservation news is not extra sales.

Complete quarterly operating model

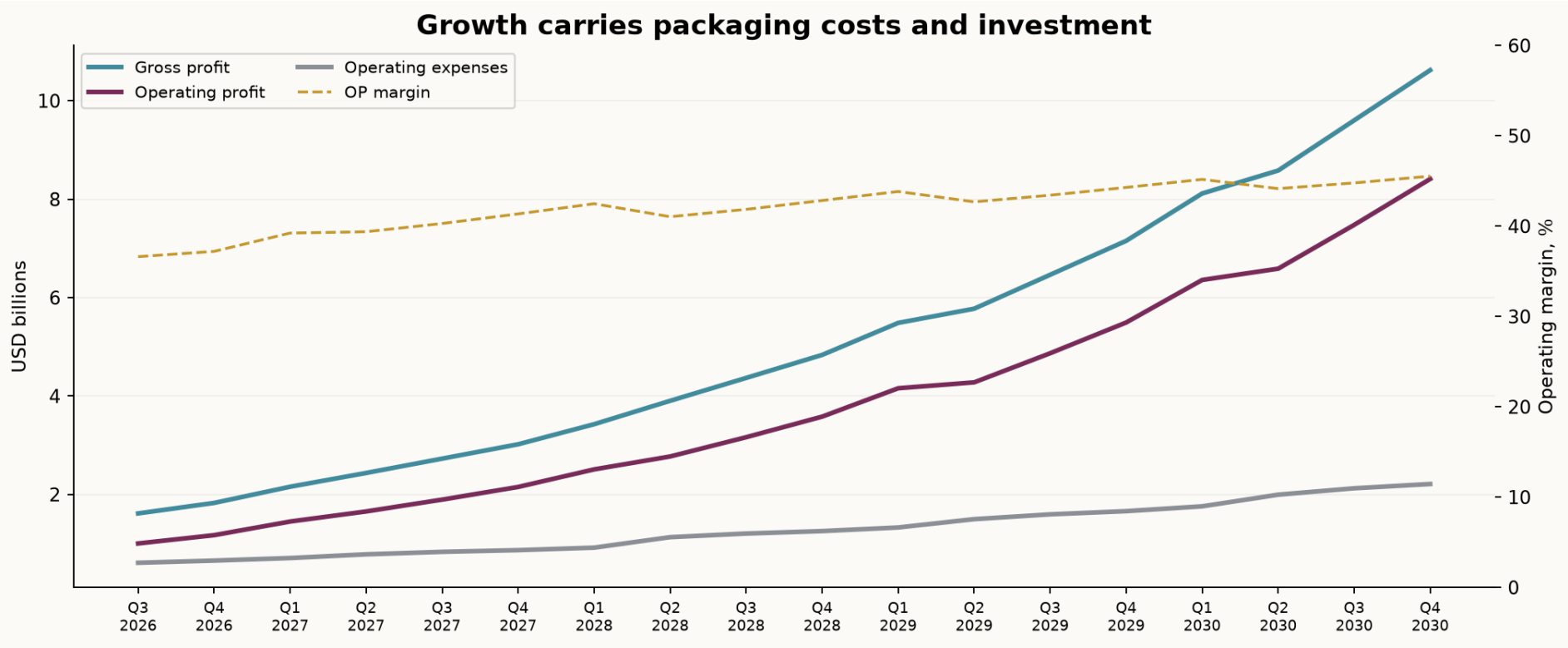
The first row is reported actual. FQ3 FY2027 uses guidance; later rows are model forecasts. Recognition follows fiscal-quarter weights of 21%, 23.5%, 26% and 29.5%, with bespoke Celestial recognition. Ordinary opex follows 23%, 24.5%, 25.5% and 27%; policy opex is added separately. These profiles are assumptions, not disclosed shipment seasonality.



01 revenue mix

Quarter	Fiscal period	Revenue, $M	GP, $M	COGS, $M
Q3 2026	FQ2 FY2027	2,739.300	1,614.000	1,125.300
Q4 2026	FQ3 FY2027	3,150.000	1,827.000	1,323.000
Q1 2027	FQ4 FY2027	3,692.900	2,156.214	1,536.686
Q2 2027	FQ1 FY2028	4,200.000	2,436.000	1,764.000
Q3 2027	FQ2 FY2028	4,701.375	2,727.306	1,974.069
Q4 2027	FQ3 FY2028	5,203.042	3,018.890	2,184.152
Q1 2028	FQ4 FY2028	5,905.177	3,426.918	2,478.259

Quarter	Fiscal period	Revenue, $M	GP, $M	COGS, $M
Q2 2028	FQ1 FY2029	6,751.601	3,902.409	2,849.192
Q3 2028	FQ2 FY2029	7,555.363	4,367.653	3,187.710
Q4 2028	FQ3 FY2029	8,359.125	4,835.245	3,523.881
Q1 2029	FQ4 FY2029	9,484.392	5,485.646	3,998.746
Q2 2029	FQ1 FY2030	10,019.428	5,774.340	4,245.088
Q3 2029	FQ2 FY2030	11,212.217	6,464.123	4,748.094
Q4 2029	FQ3 FY2030	12,405.006	7,153.906	5,251.099
Q1 2030	FQ4 FY2030	14,074.910	8,116.296	5,958.614
Q2 2030	FQ1 FY2031	14,920.300	8,582.530	6,337.770
Q3 2030	FQ2 FY2031	16,696.526	9,602.637	7,093.889
Q4 2030	FQ3 FY2031	18,472.752	10,622.744	7,850.009



02 profit costs

Quarter	Gross margin	Opex, $M	OP, $M	OP margin	Shares, M
Q3 2026	58.92%	610.800	1,003.200	36.62%	921.200
Q4 2026	58.00%	655.000	1,172.000	37.21%	921.000
Q1 2027	58.39%	707.300	1,448.914	39.24%	923.340
Q2 2027	58.00%	782.000	1,654.000	39.38%	925.680
Q3 2027	58.01%	833.069	1,894.237	40.29%	928.261
Q4 2027	58.02%	867.152	2,151.738	41.36%	930.501
Q1 2028	58.03%	918.259	2,508.660	42.48%	932.741
Q2 2028	57.80%	1,131.306	2,771.103	41.04%	940.821

Quarter	Gross margin	Opex, $M	OP, $M	OP margin	Shares, M
Q3 2028	57.81%	1,205.106	3,162.547	41.86%	948.901
Q4 2028	57.84%	1,254.319	3,580.926	42.84%	957.461
Q1 2029	57.84%	1,328.135	4,157.512	43.84%	966.021
Q2 2029	57.63%	1,497.248	4,277.092	42.69%	968.981
Q3 2029	57.65%	1,594.923	4,869.200	43.43%	972.182
Q4 2029	57.67%	1,660.061	5,493.845	44.29%	975.382
Q1 2030	57.66%	1,757.762	6,358.535	45.18%	978.822
Q2 2030	57.52%	1,994.701	6,587.829	44.15%	982.262
Q3 2030	57.51%	2,124.835	7,477.802	44.79%	985.703
Q4 2030	57.50%	2,211.624	8,411.120	45.53%	989.383

Every revenue vector and product margin

All amounts below are USD millions. Product GP is revenue times the displayed margin. Optical DSP and analog include their separate policy premium; other lines do not. Annual cost reserves flow through the displayed product margins.

Custom XPU / attach

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	650.000	55.16%	358.515
Q4 2026	900.000	50.32%	452.900
Q1 2027	1,100.000	52.00%	572.000
Q2 2027	1,470.000	50.30%	739.410
Q3 2027	1,645.000	50.30%	827.435
Q4 2027	1,820.000	50.30%	915.460
Q1 2028	2,065.000	50.30%	1,038.695
Q2 2028	2,940.000	52.00%	1,528.800
Q3 2028	3,290.000	52.00%	1,710.800
Q4 2028	3,640.000	52.00%	1,892.800
Q1 2029	4,130.000	52.00%	2,147.600
Q2 2029	4,305.000	52.00%	2,238.600
Q3 2029	4,817.500	52.00%	2,505.100
Q4 2029	5,330.000	52.00%	2,771.600
Q1 2030	6,047.500	52.00%	3,144.700
Q2 2030	6,048.000	52.00%	3,144.960
Q3 2030	6,768.000	52.00%	3,519.360
Q4 2030	7,488.000	52.00%	3,893.760

Merchant electrical

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	300.000	66.00%	198.000
Q4 2026	350.000	66.00%	231.000
Q1 2027	450.000	66.00%	297.000
Q2 2027	609.000	66.00%	401.940
Q3 2027	681.500	66.00%	449.790
Q4 2027	754.000	66.00%	497.640
Q1 2028	855.500	66.00%	564.630
Q2 2028	787.355	66.00%	519.654
Q3 2028	881.088	66.00%	581.518
Q4 2028	974.821	66.00%	643.382
Q1 2029	1,106.047	66.00%	729.991
Q2 2029	1,056.153	66.00%	697.061
Q3 2029	1,181.885	66.00%	780.044
Q4 2029	1,307.618	66.00%	863.028
Q1 2030	1,483.643	66.00%	979.205
Q2 2030	1,380.273	66.00%	910.980
Q3 2030	1,544.591	66.00%	1,019.430
Q4 2030	1,708.909	66.00%	1,127.880

External optical DSP

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	486.000	66.00%	320.760
Q4 2026	535.500	66.00%	353.430
Q1 2027	572.805	66.00%	378.051
Q2 2027	631.800	66.00%	416.988
Q3 2027	707.898	66.04%	467.469
Q4 2027	784.184	66.07%	518.129
Q1 2028	890.857	66.11%	588.931
Q2 2028	885.457	66.14%	585.679
Q3 2028	990.869	66.14%	655.403
Q4 2028	1,096.281	66.14%	725.127
Q1 2029	1,243.857	66.14%	822.740
Q2 2029	1,372.388	66.14%	907.756
Q3 2029	1,535.768	66.14%	1,015.822
Q4 2029	1,699.147	66.14%	1,123.889
Q1 2030	1,927.878	66.14%	1,275.181
Q2 2030	2,144.954	66.14%	1,418.764
Q3 2030	2,400.306	66.14%	1,587.665
Q4 2030	2,655.657	66.14%	1,756.565

External analog

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	270.000	66.00%	178.200
Q4 2026	297.500	66.00%	196.350
Q1 2027	318.225	66.00%	210.029
Q2 2027	351.000	66.00%	231.660
Q3 2027	393.277	66.04%	259.705
Q4 2027	435.658	66.07%	287.849
Q1 2028	494.920	66.11%	327.184
Q2 2028	491.921	66.14%	325.377
Q3 2028	550.483	66.14%	364.113
Q4 2028	609.045	66.14%	402.848
Q1 2029	691.032	66.14%	457.078
Q2 2029	762.438	66.14%	504.309
Q3 2029	853.204	66.14%	564.346
Q4 2029	943.971	66.14%	624.383
Q1 2030	1,071.044	66.14%	708.434
Q2 2030	1,191.641	66.14%	788.202
Q3 2030	1,333.503	66.14%	882.036
Q4 2030	1,475.365	66.14%	975.870

DCI modules

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	324.000	66.00%	213.840
Q4 2026	357.000	66.00%	235.620
Q1 2027	381.870	66.00%	252.034
Q2 2027	277.200	66.00%	182.952
Q3 2027	310.200	66.00%	204.732
Q4 2027	343.200	66.00%	226.512
Q1 2028	389.400	66.00%	257.004
Q2 2028	410.205	66.00%	270.736
Q3 2028	459.039	66.00%	302.966
Q4 2028	507.873	66.00%	335.196
Q1 2029	576.241	66.00%	380.319
Q2 2029	601.918	66.00%	397.266
Q3 2029	673.574	66.00%	444.559
Q4 2029	745.231	66.00%	491.853
Q1 2030	845.551	66.00%	558.064
Q2 2030	889.980	66.00%	587.387
Q3 2030	995.930	66.00%	657.314
Q4 2030	1,101.880	66.00%	727.241

Merchant memory

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	140.000	63.00%	88.200
Q4 2026	190.000	63.00%	119.700
Q1 2027	220.000	63.00%	138.600
Q2 2027	210.000	63.00%	132.300
Q3 2027	235.000	63.00%	148.050
Q4 2027	260.000	63.00%	163.800
Q1 2028	295.000	63.00%	185.850
Q2 2028	310.710	63.00%	195.747
Q3 2028	347.699	63.00%	219.051
Q4 2028	384.689	63.00%	242.354
Q1 2029	436.474	63.00%	274.978
Q2 2029	456.605	63.00%	287.661
Q3 2029	510.963	63.00%	321.907
Q4 2029	565.321	63.00%	356.152
Q1 2030	641.422	63.00%	404.096
Q2 2030	644.127	63.00%	405.800
Q3 2030	720.809	63.00%	454.110
Q4 2030	797.491	63.00%	502.419

Celestial

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	0.000	65.00%	0.000
Q4 2026	0.000	65.00%	0.000
Q1 2027	0.000	65.00%	0.000
Q2 2027	0.000	65.00%	0.000
Q3 2027	0.000	65.00%	0.000
Q4 2027	50.000	65.00%	32.500
Q1 2028	125.000	65.00%	81.250
Q2 2028	150.000	65.00%	97.500
Q3 2028	175.000	65.00%	113.750
Q4 2028	225.000	65.00%	146.250
Q1 2029	250.000	65.00%	162.500
Q2 2029	300.000	65.00%	195.000
Q3 2029	350.000	65.00%	227.500
Q4 2029	400.000	65.00%	260.000
Q1 2030	450.000	65.00%	292.500
Q2 2030	500.000	65.00%	325.000
Q3 2030	550.000	65.00%	357.500
Q4 2030	600.000	65.00%	390.000

Other scale-up optics

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	1.500	65.00%	0.975
Q4 2026	20.000	65.00%	13.000
Q1 2027	80.000	65.00%	52.000
Q2 2027	189.000	65.00%	122.850
Q3 2027	211.500	65.00%	137.475
Q4 2027	184.000	65.00%	119.600
Q1 2028	140.500	65.00%	91.325
Q2 2028	280.352	55.61%	155.895
Q3 2028	306.585	55.61%	170.482
Q4 2028	307.817	55.61%	171.168
Q1 2029	354.543	55.61%	197.150
Q2 2029	648.326	48.47%	314.217
Q3 2029	711.222	48.47%	344.700
Q4 2029	774.118	48.47%	375.183
Q1 2030	882.172	48.47%	427.552
Q2 2030	1,583.725	47.96%	759.517
Q3 2030	1,781.787	47.96%	854.503
Q4 2030	1,979.850	47.96%	949.489

Communications / other

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	567.800	45.00%	255.510
Q4 2026	500.000	45.00%	225.000
Q1 2027	570.000	45.00%	256.500
Q2 2027	462.000	45.00%	207.900
Q3 2027	517.000	45.00%	232.650
Q4 2027	572.000	45.00%	257.400
Q1 2028	649.000	45.00%	292.050
Q2 2028	495.600	45.00%	223.020
Q3 2028	554.600	45.00%	249.570
Q4 2028	613.600	45.00%	276.120
Q1 2029	696.200	45.00%	313.290
Q2 2029	516.600	45.00%	232.470
Q3 2029	578.100	45.00%	260.145
Q4 2029	639.600	45.00%	287.820
Q1 2030	725.700	45.00%	326.565
Q2 2030	537.600	45.00%	241.920
Q3 2030	601.600	45.00%	270.720
Q4 2030	665.600	45.00%	299.520

Valuation, dilution and analytical conviction

The sanctioned basis remains **20× forward-four-quarter non-GAAP operating profit**. The displayed ceiling quarter's share denominator, fiscal-to-calendar mapping and current rolling-horizon convention remain in force. Adjusted OP is not after-tax EPS, and the company's greater-than-USD30 FY2031 EPS aspiration does not replace this method. Recurring excluded compensation and acquisition economics remain investment risks.

Retain the saved dilution schedule: actual 921.2M and guided 921M shares anchor the path, followed by the existing 2M quarterly ordinary dilution allowance. Google warrant vesting remains linked to the explicit qualifying-revenue subset already inside custom. Its maximum 58.970907M shares comprises 1.360867M time-based

shares and 57.61004M purchase-based shares across 240 USD500M tranches. The USD120B opportunity is not committed backlog. Full vested shares are the model's price-independent proxy, not GAAP treasury-stock dilution. [M2; M3]

Celestial's existing full contingent-share assumption remains 5.6M/11.2M/16.8M/22.4M cumulative across FY2029, retained thereafter. That is a conservative modeled issuance path, not verified milestone dates. Upfront acquisition shares are already in the current count. The cash-settled hedge is not a share repurchase. Capacity or pricing revisions do not by themselves authorize a different share schedule. [M3; T5]

Analytical category	Score	Current reasoning
Core Thesis	19/20	Broad qualified IP and custom relationships address multiple constraints in data movement.
Cycle Stage	10/10	Actual production and rising near-term guidance establish an accelerating ramp.
TAM Expansion	10/10	Attach, scale-up and memory architectures expand the addressable wallet.
Catalysts	5/5	Accepted ramps, faster optics, switch launches and quarterly execution test the Investor Day plan.
Macro	9/10	AI network and memory intensity support demand, with correlated customer spending risk.
Alpha	3/5	Program mix and warrant economics matter, while major relationships are public.
Risks	-5	Customer concentration, competition, supply/qualification, integration and adjusted-profit/dilution economics.
Total	51/60 = 85/100	Analytical Base retained.

Protected portfolio Excel Base: 85/100. Artifact analytical Base: 85/100. Neither is a contract probability or a second multiplier on the price ceiling. The new operating evidence enters the financial bridge rather than an automatic score or multiple increase.

Long-term Non-GAAP financial model

Metric	FY31 target model
Revenue	$70-90B
Gross Margin	56-59%
Operating Margin	44-46%
Tax Rate	15%
Free Cash Flow	>36%



FY31 implied EPS

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Marvell Investor Day, PDF page 142. FY2031 management targets include 56-59% non-GAAP gross margin, 44-46% operating margin, 15% tax and free-cash-flow margin above 36%. EPS and cash-flow targets do not replace the model's operating-profit valuation basis.

Effects on suppliers and competing architectures

The strongest implications extend beyond Marvell's own revenue. More optical connections, larger packages and shared memory can increase spending on qualified light sources, photonic wafers, substrates and test equipment. At the same time, Marvell's complete engines and modules can displace external component or module invoices. The relevant question is which company owns the incremental qualified product and retains profit after manufacturing and development costs. These are transmission mechanisms and research priorities, not supplier awards inferred from an architecture diagram.

Qualified light sources and the rest of the optical assembly

Lumentum (LITE) has the clearest product fit with the qualified-light requirement. Its March OFC release describes high-power, narrow-linewidth sources for direct-detect and coherent-lite links. The named Marvell-Lumentum OCS interoperability demonstration is a separate, real ecosystem relationship; it does not identify the laser supplier inside every Marvell engine. More optical bandwidth can increase demand for usable light, but splitting, wavelength multiplexing, coupling losses and laser count per engine determine supplier revenue. The current LITE model already includes its UHP ramp and a 55%-weighted scarcity-pricing case. The next useful evidence is allocation, accepted source output and net ASP beyond that existing path, with burn-in, coupling, yield and support costs included. [SO1; SO2; DECK pp67,84,95,111]

Coherent (COHR) has a broader opportunity than the high-power laser comparison alone. Its PhotonLink portfolio spans light generation, receiving, fiber and integration. Detector, optical-frontend and assembly content can grow even if a particular laser product has qualification disadvantages. Conversely, Marvell's complete coherent modules can compete with module revenue elsewhere. The existing 25% UHP risk remains scoped to the exposed product pool; the deck neither removes that risk nor turns it into a company-wide failure thesis. **Sivers (SIVE.ST)** and **Aeva (AEVA)** have potentially relevant source technologies, but the slides name neither as the supplier. A similar 2027 roadmap is insufficient to identify Aeva's undisclosed partner. [SO3; DECK pp94-95,111]

Photonic foundries and optical testing

Tower (TSEM) and **GlobalFoundries (GFS)** have named Marvell relationships. Tower's June 18 release reports more than five million coherent PICs shipped through the collaboration. That establishes a commercial route in coherent connectivity, not an allocation of every new CPO engine; cumulative shipments are not an annual sales forecast. GF's September SiGe capacity agreement supplies a different part of the optical chain. More complex and faster optical links can increase their qualified wafer demand, but utilization, die area, good-unit yields, agreed wafer prices and customer acceptance determine profit. A GF agreement is not evidence that Tower loses an unrelated process or product. Both relationships already appear in current company research. [SO4; M5]

Chroma (2360 TW) is a particularly useful second-order follow-up. Optical engines add test and alignment challenges as electrical, photonic and packaging functions become more tightly integrated. Our current Chroma research records received CPO-related orders and management's July call discussion that Marvell had not yet ramped. Its model already applies 55% to an incremental CPO wallet after subtracting embedded semiconductor growth. Marvell's roadmap makes customer conversion worth tracking; it is not a second order book. The next bridge is customer tool orders, test time/content, shipment and acceptance timing, then gross profit after qualification/R&D and competing tool share. [SO5; DECK pp71,111,125]

Substrates and the equipment behind them

AT&S (ATS, Vienna) is a direct named route through Marvell's September capacity agreement and is already a research candidate on the To do list. **Samsung Electro-Mechanics (009150)** has a broader substrate read-through, while **Taesung (323280)** supplies equipment further upstream. The deck's increasingly complex packages can require larger substrates and more demanding processing. That can reduce good-unit output per line, improve the value of qualified supply and encourage capacity spending. It does not establish that every substrate vendor receives the same pricing premium or that Taesung books Marvell's package spending as equipment revenue. [M4; DECK pp71,125]

For substrate makers, the bridge is qualified output times utilization and realized price, less yield loss, materials, depreciation and startup costs. For Taesung, it is the relevant wet-process equipment wallet, vendor share and accepted deliveries, with installation and warranty costs. Existing AT&S/core orders, SEMCO expansion assumptions and Taesung's 45% pricing outcome are already represented in their respective research or backlog. The useful catalysts are new allocations, repriced orders and customer acceptance beyond those paths. **TSMC (TSM)** can also benefit from advanced-node and packaging demand, but the deck alone does not isolate incremental sales beyond its existing AI mix.

Merchant components and modules face a mixed outcome

Semtech (SMTC) can benefit from more high-speed driver and amplifier content, and its 200G XPO products provide a real merchant route. Marvell also designs analog parts and sells integrated engines. Customer choice between merchant components and a Marvell assembly determines whose invoice survives. The current Semtech capacity path already includes the Tower expansion hypothesis; the next task is a customer/SKU capacity-to-revenue reconciliation, not applying the whole optical TAM to Semtech. [SO6; DECK pp77-78,94-95]

AAOI benefits from near-term 800G/1.6T pluggable demand, while NPO/CPO can shift part of the later wallet away from pluggables. **Fabrinet (FN)** may gain difficult optical assembly work if customers outsource it, but material pass-through can fall as the architecture changes, and the deck announces no new Fabrinet award. **Credo (CRDO)** has optical products as well as copper cables, so it is not simply a loser from optical adoption; link reach, topology and accepted program wins determine substitution. **Astera Labs (ALAB)** has memory-expansion and connectivity exposure, alongside competition from Marvell's controllers, switches and fabric. **Broadcom (AVGO)** remains a major competitor. A larger market does not mean all competing vendors can count the same socket.

Shared memory changes the mix rather than removing the memory requirement

For **Micron, SK hynix and Samsung**, memory pooling can lower the amount of expensive local memory needed for a given workload while creating demand for another memory tier. Marvell's August Photonic Fabric description is specific: a PFMM supports up to eight DDR5 DIMMs, 72GB of HBM cache and 7.2Tbps of optical-fabric bandwidth. It is not an HBM-free system. At the transcript's illustrative eight-to-sixteen modules per appliance, that configuration would contain 576-1,152GB of HBM cache and up to 64-128 DDR5 DIMMs; these are conditional content calculations, not forecast shipments or net incremental memory demand. [SO7; ID287-289; DECK p121]

DDR4 reuse belongs to separate CXL controller paths, not this DDR5 PFMM configuration. The economic outcome combines displaced local HBM, new shared DDR5/HBM content, controller and optical costs, and additional workloads enabled by lower system cost. Existing memory research already carries a 25% probability for material optical-attached DRAM adoption by end-2029, including DRAM recapture and cost offsets. The deck therefore warrants neither a fresh blanket memory uplift nor a duplicate HBM haircut. Measured deployment economics and changes in supplier mix are the next evidence to seek.

Research priority: LITE for qualified source allocations and pricing; Chroma for Marvell-linked test conversion; Tower for accepted wafer output and product allocation. Keep SMTC's merchant-versus-integrated capture check prominent, and AT&S as the direct substrate candidate already in the backlog. These are priorities

for resolving company economics, not new Base scores or target changes. No additional Marvell revenue or profit is booked from these supplier implications, and no other company's model is edited in this assessment.

Second-order sources: SO1: Lumentum OFC product disclosure, March 17, 2026; SO2: Marvell-Lumentum OCS demonstration, March 16, 2026; SO3: Coherent PhotonLink; SO4: Tower-Marvell coherent PICs, June 18, 2026; SO5: Chroma Q2 presentation and the management-call record in the current Chroma artifact, July 30, 2026; SO6: Semtech 200G XPO analog products, September 21, 2026; SO7: Marvell Photonic Fabric technical description, August 4, 2026. Existing company-model overlap is assessed against the current research artifacts as of October 6, 2026. M4 and M5 below identify the AT&S and GF agreements.

Risks and dated tests

Risk or catalyst	What would strengthen the case	What would weaken it
Next quarterly report; December 2026 monitoring window, actual date to be confirmed	Delivery against USD3.15B FQ3 guidance and evidence supporting FY2028 accepted demand	Deferred acceptance, weaker mix or costs exceeding the operating bridge
FY2028 ramps, February 2027-January 2028	Scale-out, scale-up and switching sales reconcile to company guidance without duplicate invoices	Pilot activity mistaken for volume; 1.6T adoption or qualification slips
T100 and U115, 2027	Named production deployments and demonstrated power/latency economics	Broadcom or other alternatives win programs; large-die yield or software delays
Celestial FY2028 exit, January 2028	Shipments support the previously stated USD500M annualized exit rate	Annualized run rate confused with full-year sales or contingent issuance outruns delivery
Custom FY2029, February 2028-January 2029	Multiple programs substantiate >USD12B guidance and the model's higher capture assumptions	Lower attach content, customer insourcing or program concentration
Foundry, substrate and optical capacity; each report	Qualified output, acceptable yields and contractual allocation support shipments	Prepayments precede excess inventory, or packaging/test bottlenecks constrain saleable units
Customer forecasting; continuous	Customer deployment and utilization validate orders	Correlated double-ordering, power/site delay or capex retrenchment
Scale-in; 2027-2030 development checks	Qualified systems, net content and production economics emerge	Demonstrations fail yield, reliability, fiber-assembly or economic tests

Source interpretation and remaining ambiguities

The supplied transcript and all 161 deck pages were reviewed. The deck repeats the absolute FY2029 custom target ($12B+) and FY2031 target ($30B+), but page 28 also repeats >3× FY2029 versus FY2028. The current $7B/$14B custom allocation does not satisfy that ratio; the exact FY2028 base and program scope remain unresolved. Neither the year nor the ratio has been silently repaired.

The visual source resolves the $1.6T label: it is CY2030 accelerated-compute TAM (page 29), not current-year data-center capex. The deck separately shows approximately $3.1T CY2030 total data-center capex and approximately $1.2T CY2026. These market figures are context, not a direct multiplier for company revenue.

Two slide errors are controlled explicitly. Page 32 retains a stale 50% growth label while showing $20B FY2028; final page 33 correctly shows 67% on $12B FY2027. Page 152 labels its $1,003.2M bottom row operating expenses, but it is non-GAAP operating profit: $1,614.0M gross profit less $610.8M opex, also equal to $459.7M GAAP OP plus $543.5M adjustments. The active model already uses the correct arithmetic.

The 100% A0-to-production statement is a design-execution metric, not 100% wafer yield. Node-readiness stages, product demonstrations and accepted sales remain distinct; existing yield and packaging cost reserves are retained. [DECK pp28-33,47,62,81,84,152]

Other normalization limits: Celestial completed in February 2026, not the transcript's 'last year'; 'AI RAM' is consistent with an AI-RAN transcription; 'SPATs' is inconsistent with Fast Pipe in adjacent sentences; the repeated micro-VCSEL wording is clarified elsewhere by micro-LED/micro-VCSEL. No financial increment depends on these wording repairs.

Technical rhetoric also needs scope. vLLM's 2023 contribution was PagedAttention, not invention of KV caching. Faithful eviction and reload/recomputation primarily impose performance costs; hallucinations are not an automatic consequence. The 85% traffic share, 15× bandwidth comparison and relative power/latency claims remain management descriptions without a matched independent benchmark. They explain motivation but do not supply exact revenue or margin multipliers. [T9]

Sources

- **ID:** User-supplied Marvell Investor Day transcript, October 6, 2026, all 519 numbered lines reviewed. ID references use transcript line numbers.
- **DECK:** User-supplied marvell-investor-day-2026-presentaion-deck.pdf, October 6, 2026, all 161 PDF pages reviewed with charts and footnotes. DECK references use one-based PDF page numbers; some internal slide footers differ. Original source visuals are reproduced intact. Official event page.
- **M1:** FQ2 FY2027 release, August 27, 2026.
- **M2:** Google agreement and warrant 8-K, August 19, 2026.
- **M3:** FQ2 FY2027 10-Q, August 28, 2026.
- **M4:** AT&S-Marvell capacity agreement, September 22, 2026.
- **M5:** GlobalFoundries-Marvell SiGe capacity agreement, September 17, 2026.
- **T1:** Marvell 1.6T DSP portfolio, March 12, 2026.
- **T2:** Marvell ECOC 2nm demonstrations, September 20, 2026.
- **T3:** Teralynx T100, June 1, 2026.
- **T4:** Broadcom Tomahawk 6 production, March 12, 2026.
- **T5:** Celestial acquisition completion, February 2, 2026.
- **T6:** Marvell custom SRAM, June 17, 2025; original saved Hot Chips image and Irrational Analysis June 2 article reviewed in the preceding research batch.
- **T7:** Structera product announcement, July 30, 2024; DDR4 reuse explanation; Vera interoperability, September 9, 2026.
- **T8:** Marvell-Mojo Vision development agreement, March 12, 2026.
- **T9:** PagedAttention original paper, September 2023; vLLM introduction, June 2023.

Reproducibility and source limits

The complete supplied transcript, Q&A and 161-page deck were reviewed. Transcript SHA-256: 2512973f606b18e697d0f9a902ca260efd289cd14c127986c1e67a3645bbe6d1. Deck SHA-256: 1e6bc0130675f91e1f5ec9ef2e27e83df7a70658d815244c2f2db4a07b52a340. Original sources and recoverable prior artifacts are retained. The official Investor Day event page confirms the event; the supplied PDF is the direct source for its visuals.

Deck integration is an evidence and visual update. It has 100% event overlap with the October 6 transcript: no additional demand, price, volume or margin increment is added. The existing 70% probability on the bounded $70-90B FY2031 opportunity remains; the $80B conditional midpoint, cost bridge, $70.970B active revenue and share schedule are unchanged. Near-term $20B guidance is already modeled. Analytical Base remains 85, protected Excel Base 85 and the Q4 2027 production ceiling $227.7064 at 20×. The separate X article retains the authorized 30-35× presentation ($341.56-$398.49).

Material remaining assumptions are the mix between merchant components and integrated engines, net prices, qualified capacity, yields, customer acceptance timing, Google warrant revenue accounting and detailed custom program allocations. The model has a company-guidance and product-mix bridge rather than independently verified physical shipments for every vector. Cash conversion also depends on capacity deposits, inventory and acquisitions. No unquantified cash balance is substituted into the preserved operating-profit valuation.

The active horizon remains Q4 2027 through December 31, 2026. The next scheduled convention roll is January 1, 2027 to Q1 2028; it is separate from an economic upgrade. Workbook locations: MRVL_Q_Model, Master Portfolio row 54 and Quarterly Ceilings row 57. The 18 operating quarters, 15 ceiling quarters, policy bridge and protected Base values are reconciled to the saved workbook.